

TECHNOPRENEURSHIP

COV-ENT121

COVENANT UNIVERSITY

CENTRE FOR ENTREPRENEURSHIP
AND DEVELOPMENT STUDIES



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TECH

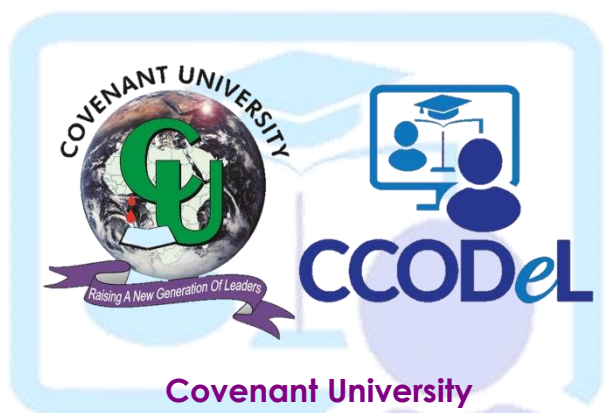
*Covenant Centre for Open and
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Course Manual Designed by



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FOREWORD

Presenting this Course Manual from the Department of Computer and Information Science on the Platform of the Covenant Centre for Open, Distance and eLearning (CCODEL), Covenant University, for our learners is a dream come true. Over the last few years, the university has made concerted efforts to join the League of Distance Education, which runs dual-mode universities by developing interactive and engaging course materials. This has become a reality. I, therefore, start by congratulating you and welcoming you to the Open, Distance and eLearning Mode of Covenant University. This online mode is not the same quality as the face-to-face mode in the university. Course material is of high quality and has all the trappings of a seamless approach to learning. It is a lecture platform in your hand.

The modules in this Course Material are written conversationally and well-structured to enhance your understanding of the course. There is also a study guide, in-text questions, and additional reading materials to expose your understanding of the pedagogy of this course. In addition to the rich academic contents of the course materials you are expected to study, you are also expected to engage in collaborative and self-learning activities that are well-structured in the Course Text Material. The stimulating academic activities in the book are what distinguish you as a self-paced learner who should see the Course Text Material as a mobile classroom.

I heartily commend this course material as a delightful study experience through which you will seamlessly enjoy your programme at this university. Our university prides itself on excellence both in morals and in knowledge. You are, therefore, admonished to distinguish yourself as a good ambassador of Covenant University.

Congratulations, and welcome on board!

Prof. Abiodun H. Adebayo

Vice-Chancellor



About this Course Manual



This course content is developed, localised, and provided to you by the Covenant Centre for Open and Distance e-Learning (CCODEL).

Course Overview

Course Code: COV-ENT121

Course Title: Technology Entrepreneurship
(Technopreneurship)

Credit: 1 Credit Unit

Year: One

Semester: Alpha

Facilitator's Information: Engr. Tola Akinyeoluwa

Phone: 08173081045

E-mail: tola.akinyeoluwa@covenantuniversity.edu.ng

Office Hours: Virtually any time by appointment. Please e-mail me with any questions, and I'll respond within 24 hours, usually sooner.



For further information or questions about the course, reach out to the contact person above.

Texts and Course Materials:

The major components of the course are the Course Guide, Study Units, Textbooks, Assignment File, and Presentation Schedule.

Recommended Textbooks:

- **Life in Media by Mark Deuze**

A Global Introduction to Media Studies

ISBN: 9780262545587

Publisher: The MIT Press

Pub Date: July 25, 2023

- **Inventing Future Cities by Michael Batty**

ISBN: 9780262548656

Publisher: The MIT Press

Pub Date: May 7, 2024

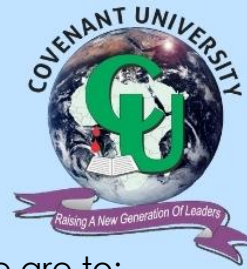
- **Prison Media by Anne Kaun and Fredrik Stiernstedt**

Incarceration and the Infrastructures of Work and Technology

ISBN: 9780262545495

Publisher: The MIT Press

Pub Date: May 9, 2023



1.0 Course Description

The specific objectives of the course are to:

1. Discuss the meaning of technopreneurship by identifying various components of technopreneurship
2. Demonstrate how to deploy social media for entrepreneurial marketing;
3. Illustrate how content marketing is created and applied in business;
4. Describe how to abide by email marketing best practices;
5. Demonstrate how to develop a sales page;
6. Illustrate how to identify various forms of cyber threats, vulnerabilities and attacks;
7. Illustrate the art of protecting business secrets.



TECHNOPRENEURSHIP

COV-ENT121

2.0 Course Learning Outcomes

After completing this course, the learner ought to be capable of:

1. Use the right technology for entrepreneurial endeavours following the functions of the various technopreneurship components.
2. Deploy different social media marketing strategies for entrepreneurial success.
3. Create suitable content marketing for an identified business
4. Deploy email marketing for entrepreneurial success and abide by the best practices.
5. Develop professional sales pages for their businesses or other identified businesses.
6. Detect various forms of cyber threats, vulnerabilities and attacks.
7. Demonstrate how to protect business secrets.

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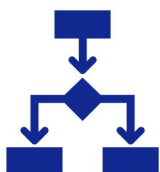
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3.0 Activities to Meet Learning Outcomes

Technopreneurship development combines technology innovation with entrepreneurial skills to create market-

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responsive products or services, particularly in social media marketing and **cybersecurity**. Several activities align with this role:

- The focus is on creating innovative social media marketing tools
- The focus is on developing **cybersecurity** solutions for social media platforms.
- The text is about the launch of educational initiatives.
- The article discusses the development of secure social media management platforms.
- Engage in consultancy services
- Collaborating with social media platforms
- Developing safe payment methods for Online Sales
- Research and development

By engaging in these activities, technopreneurs can play a crucial role in driving innovation and development in social media marketing and **cybersecurity**, ultimately contributing to the growth and sustainability of businesses operating in the digital realm.



3.0.1 Group Assignments

You are expected to work together on team assignments. This mimics the typical team-based nature of our work environment. Please remember the basics of group dynamics (Forming, Storming, Norming and Performing) and try to quickly adjust in your teams so you can begin to perform for success.

For a comprehensive understanding of the material, keeping current with the reading is strongly encouraged. Assignments on the course syllabus are to be completed within the week. Late assignments will not be accepted.



3.0.2 Personal Introduction

For the first assignment, please prepare a short biography that includes your current work status, experience, and any other information you'd like to share.

3.0.3 Group Discussion Forum



The weekly assignments are listed in the Course Outline below. Submit your group's response each week they are due.

3.0.3.1 Discussion Post and Responses

Prepare to post your original thoughts on each forum topic and respond to two other classmates. Sharing WHY your opinion is the same or differs is the key to an enlightening discussion. Indicate whose post you are responding to and what specific points you are addressing. Also, bringing in additional outside resources, knowledge, and related life experiences would enhance your response and command additional points. Please pay attention to the discussion board grading rubric below.

3.0.3.2 Discussion Board Grading Rubric

The discussion board grading rubric gives students guidelines for completing discussion forum assignments. It shows the grading criteria for the assignment.

DISCUSSION BOARD GRADING RUBRIC

STUDENT:

Criterion	Point Value	Points Possible	Points Earned & Instructor Comments
Thread			
All key components of the Discussion Board question are answered in a new thread.	10	30	
Clear, logical flow to post	2.5		
Major points are stated clearly.	2.5		
Major points are supported by the following: Textbook or articles (two citations)	10		
Good examples (pertinent, conceptual, or personal examples are acceptable)			
Thoughtful analysis (considering assumptions, analysing implications, comparing/contrasting concepts)			
Follows APA format			
Spelling & grammatical-error-free	2.5		
Sufficient word count (Minimum 300 words)	2.5		
Two replies:	5	20	
Sufficient word count (Minimum 150 words)			
Major points are supported by the following: Textbook or articles (one citation)	10		
Good examples (pertinent, conceptual, or personal examples are acceptable)			
Thoughtful analysis (considering assumptions,			



analysing implications, comparing/contrasting concepts) Follows APA format.			
Appropriate “netiquette” manners (For example, no name calling or labelling another student’s idea in a derogatory term, such as “stupid” or “dumb” even when disagreeing – See Student Expectations)	2.5		
Spelling & grammatical-error-free	2.5		
Total		50	

3.0.4 Discussion Board Questions



1. Due at the end of Week 3: Please post your comments on the discussion board. The use of additional resources and research will attract additional marks. **Submit your written document in the assigned Cloud Storage.**

The midterm and final exams are both multiple-choice quizzes to be completed online.

3.0.5 Course Technology



There are video recordings and PowerPoint presentations for each week. Please take the time to go through the lecture materials.

4.0 Evaluation

The final grade will be calculated by using the following items and weights:

12 Individual Discussion Assignments/Responses (TMA)	400 points
1 Final Exam	600 points
Total Possible	1,000 points

5.0 Grading Scale

The following is the grading scale:

Score %	Letter Grade	Grade Point	Remark
70 and above	A	5	Pass
60-69	B	4	Pass
50-59	C	3	Pass
45-49	D	2	Pass
0-44	F	0	Fail



6.0 Academic Rules

6.0.1 Course Work

All assignments must be completed by the due date.



6.0.2 Academic Integrity

Students are encouraged and expected to conduct themselves following the highest standards of academic honesty. It is assumed and expected in a learning community that academic honesty is practised, that the work of others is acknowledged and cited where required, and that the work submitted for credit, as individuals or team members, is one's work. Any student who engages in academic misconduct during this course will, at a minimum, receive a failing grade. Please consult the CU's Academic Handbook for the consequences of Academic Dishonesty.

6.0.3 Participative Learning

I encourage you to have management information system practice at the back of your mind as you participate in this course. Please familiarise yourself with course assignments. Work closely with your teams.



7.0 Open Education Resources



Open Education Resources are books and publications that have been placed in the public domain for people in the learning community. These are usually free if they are not offered for resale. As you progress in your studies and do research work, you can make use of these. Following are examples of Open Education Resources. You can find many more online.

Books

1. Unlocking the potential of youth entrepreneurship in developing countries: From Subsistence to Performance (2017). Paris: Organisation for Economic Cooperation and Development (OECD). Retrieved from <https://www.proquest.com/books/unlocking-potential-youth-entrepreneurship/docview/1924617506/se-2>



2. Gandhi, S. J., & Sharma, N. (2016). Innovation and entrepreneurship. 2nd edition; engineering management handbook, 2nd edition, Huntsville (2nd edition ed., pp. 379-392). Huntsville: American Society for Engineering Management (ASEM). Retrieved from <https://www.proquest.com/books/innovation-entrepreneurship/docview/2173845017/se-2>
3. Kariv, D. (2011). Entrepreneurship: An international introduction. Florence, Florence: Taylor & Francis Group, Taylor & Francis Group. Retrieved from <https://www.proquest.com/books/entrepreneurship/docview/2130966303/se-2>
4. Nordic Council, o. M. (2015). Social entrepreneurship and social innovation in the Nordic countries: Initiatives to promote social entrepreneurship and social innovation. Copenhagen, Copenhagen: Nordic Council of Ministers, Nordic Council of Ministers. Retrieved from <https://www.proquest.com/books/social-entrepreneurship-innovation-nordic/docview/2148271288/se-2>

Journals

1. AlOmari, A. M. H. (2024). Game theory in entrepreneurship: A review of the literature. Journal of Business and Socio-Economic Development, 4(1), 81-94. <https://doi.org/10.1108/JBSED-01-2023-0005>
2. Maureen, S. A., Benson, D., Schill, R., Ronald, M. M., & Schill, A. (2024). Drivers of entrepreneurship in Latin America: Income, education, and work status. The International Journal of Interdisciplinary Social and Community Studies, 19(1), 83-105. <https://doi.org/10.18848/2324-7576/CGP/v19i01/83-105>
3. Minja, J. J., Charles, G., & Mbura, O. K. (2023). The influence of co-production of entrepreneurship training programmes on the creative transfer of entrepreneurship skills. Journal of Global Entrepreneurship Research, 13(1), 4. <https://doi.org/10.1007/s40497-023-00349-1>
4. Pašić, M., Vatreš, A., & Bijelonja, I. (2023). Analysis of the development of entrepreneurship competencies of engineering students based on the EntreComp framework. International Journal of Industrial Engineering and Management, 14(2), 162-182. <https://doi.org/10.24867/IJIE-2023-2-331>



5. Whitaker, A. (2023). The rise of hybrid practice: Creative institutional design as arts entrepreneurship. *Artivate*, 11(3)
<https://doi.org/10.34053/artivate.11.3.192>

8.0 Other Sources for Open Education Resources

8.0.1 Universities

- [The University of Cambridge's Guide on Computer Network Security](#)
- [The University of Cambridge's Guide on Cybersecurity Collection at Galileo](#)
- [The University of Cambridge's Guide on Ethical Hacking for Effective Defence \(Modules, Labs, and Lectures\)](#)
- [Open Learn](#) from Open University in the UK

8.0.2 Global

- Enesco's [searchable open database](#) is a portal to worldwide courses and research initiatives.
- African Virtual University (<http://oer.avu.org/>) has numerous modules on subjects in English, French, and Portuguese.
- <https://code.google.com/p/course-builder/> is Google's open-source software for anyone creating online education courses.
- Global Voices (<http://globalvoicesonline.org/>) is an international community of bloggers who report on blogs and citizen media worldwide, including open-source and open educational resources.

8.0.3 Individuals (which include OERs)

- [Librarian Chick](#): Everything from books to quizzes and videos here, including directories on open source and open educational resources.
- [K-12 Tech Tools](#): OERs, from art to special education.
- [Web 2.0: Cool Tools for Schools](#): audio and video tools.
- [Web 2.0 Guru](#): animation and various collections of free, open-source software
- [Live binders](#): search, create, or organise digital information binders by age, grade, or subject (why reinvent the wheel?).



8.0.4 Legal Help

[New Media Rights](#) is trying to help digital creators use public domain or open materials legally. They have guides on using free and open software materials in various fields.

9.0 Course Outline and Activities

WEEK/DATE	MODULE	UNIT TOPICS	TEAM ACTIVITIES	INDIVIDUAL ACTIVITIES	REMARKS
WEEK ZERO		None	None	None	Review the entire course site
WEEK 1	MODULE 1	Unit 1: THE ROLES OF TECHNOLOGY IN ENTREPRENEURIAL DEVELOPMENT • Introduction • Learning Outcomes • Overview Of Technopreneurship • Synopsis Of Technology-Driven Business • Importance Of Technopreneurship • Technopreneurship Processes • Technopreneurship Examples • An Illustration Of a Prosperous Technopreneur • A Successful Technopreneur's Attributes	Team formation: Form teams of 3-4 members each. Topic Selection: within the intersection of entrepreneurship, social media, and cybersecurity	1. Complete assignment: Team formation 2. Complete assignment: Technology Analysis	Remember to submit your teamwork assignment within a week.
WEEK 2	MODULE 2	Unit 2: SOCIAL MEDIA MARKETING • The Roles Of Technology In Social Media Marketing • Social Media's Development And History • Use Of Social Media For Entrepreneurial Marketing • Why Use Social Media For Business?	Market Research: Teams research their chosen topic.	Complete Assignment: Research and analysis	The deadline for posting original thoughts is Wednesday this week, and responses are encouraged.
WEEK 3	MODULE 3	Unit 3: LEADING SOCIAL MEDIA NETWORKS FOR BUSINESSES • How To Utilize Social	Proposal Development: Based on their	Complete assignment: Proposal	All team assignments should be



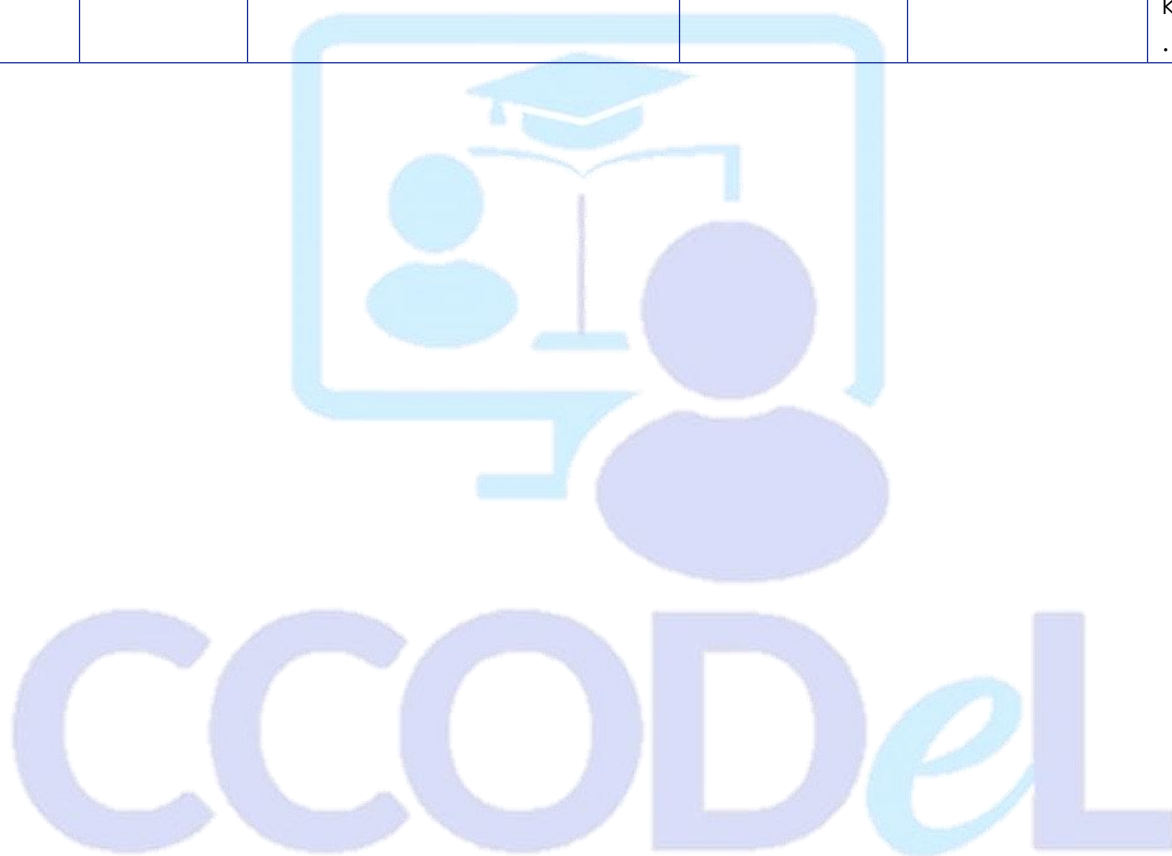
		Media At Work • Using Facebook, Instagram, WhatsApp, Messenger, And Meta For Business • Business Suggestions For Social Media • Social Media Automation: What Is It?	research, teams develop a comprehensive proposal	development	submitted to the drop box.
WEEK 4	MODULE 4	Unit 4: BEST SOCIAL NETWORKING SITES FOR COMPANIES AND PLATFORMS • Benefits And Drawbacks Of Social Media Marketing (SMM) Benefits • Negative Aspects	Presentation Preparation: Teams prepare a presentation to communicate their proposal effectively	Complete assignment: Presentation preparation	The success of the exercise and actions should be documented and submitted to DropBox.
WEEK 5 & WEEK 6	MODULE 5	Unit 5: CYBERSECURITY • Cybercriminals Against Experts In Cybersecurity • Why Become a Cybersecurity Expert? • Typical Dangers • Increasing Risks To Cybersecurity • Developing Additional Experts To Solve The Cybersecurity Expert Shortage	None		



WEEK 5 & WEEK 6	MODULE 6	Unit 6: THE CIA TRINITY: AVAILABILITY, INTEGRITY, AND CONFIDENTIALITY • Attacks, Vulnerabilities, And Threats To Cybersecurity • Guarding Against Trojan Horses • Protecting Against Browser And Email Attacks • Protecting Against Attacks: How To Keep Information Safe	Presentation and Peer Review: Each team presents their proposal to the class	Complete assignment: Presentations and peer review	The deadline for posting original thoughts is Wednesday this week, and responses are encouraged.
WEEK 7	MODULE 7	Unit 7: CONTROLS FOR PHYSICAL ACCESS • Authentication with many factors • Control mechanisms • The craft of maintaining honesty	Final Presentation and Peer Review: Each team presents their proposal to the class. After each presentation, other teams provide constructive feedback and ask questions to deepen their understanding of the topic.	Complete assignment: Presentations and peer review	
WEEK 8	MODULE 8	Unit 8: THE CODE OF ETHICS FOR CYBERSECURITY EXPERTS • Hashing: what is it? • A digital signature: what is it? • The idea of the five nines	Submission of final report: Teams submit a final written report that documents their proposal in detail	Complete assignment: Submission of final report	The exercise's success and actions should be documented and



		• Creating a High Availability System • Systems for preventing intrusions			submitted to DropBox, ensuring you have developed sufficient capacity to effectively impart your knowledge .
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10.0 MODULE 1: THE ROLES OF TECHNOLOGY IN ENTREPRENEURIAL DEVELOPMENT

10.0.1 UNIT 1: THE ROLES OF TECHNOLOGY IN ENTREPRENEURIAL DEVELOPMENT

1.0 Introduction



This course will help to equip Covenant University graduates with the contemporary skills needed for modern-day entrepreneurship. It further emphasises the need for the deployment of technology in entrepreneurial processes.

Apart from deploying technology for ideation and start-up formation, this course shall focus on **cybersecurity** and social media marketing. Cyber security is about protecting yourself and others from attacks carried out primarily with computers. Meanwhile, social media marketing entails the deployment of social media for marketing activities.

2.0 Learning Outcomes

After completing this course, the learner ought to be capable of:



- use the right technology for entrepreneurial endeavours by the functions of the various technopreneurship components
- deploy different social media marketing strategies for entrepreneurial success
- create suitable content marketing for an identified business
- deploy email marketing for entrepreneurial success and abide by the best practices
- Develop professional sales pages for their businesses or other identified businesses
- detect various forms of cyber threats, vulnerabilities and attacks; and
- demonstrate how to protect business secrets.

3.0 Overview of Technopreneurship





Technopreneurship is a process of leveraging technology to find novel ideas.

To put it briefly, technopreneurship is leveraging technology to find novel ideas. It is applying technology to progress a nation or the entire world.

The components of this technopreneurship course are social media marketing and **cybersecurity**. This forms the content of this course.

3.0.1 Synopsis of Technology-Driven Businesses

A **BUSINESS-SAVVY** individual can use technology to establish, grow, and profitably generate a profit through a process known as technopreneurship. The first phase in the process is to identify a business opportunity. Next, a product or service that capitalises on the opportunity to maximise profit is developed. [Go to this URL for more information.](#)

In the Centre for Entrepreneurial Development Studies (CEDs), for example, Covenant University's brand of digital technology training on innovation, creativity, and productivity currently offers Cisco certification programs with a focus on:

- Web Development - JavaScript
- Python - Python Essentials
- **IOT** - **IOT** connecting things
- **Cybersecurity** - **Cybersecurity** Essentials
- Graphics - Get connected
- Cinematography - Get connected
- Solar panel - CCNA1
- Phone/Laptop repairs - IT essentials

The aim of the several ICT programs is for students to acquire the essential skills and develop Internet of Things (**IoT**) solutions related to our ecosystem for [more information](#).

As a result, technopreneurs must be problem solvers, creative product and service developers, and business managers of their enterprises' finances and operations. Since technopreneurship is a type of entrepreneurship, a comparison with entrepreneurship can help us understand it better.

Technopreneurship And Entrepreneurship Share Similarities.



Technopreneurs and entrepreneurs alike need to be innovative thinkers to stand out in the market and offer value to their clientele.

Technopreneurship and entrepreneurship share a common trait: they both involve starting and growing a new business enterprise. The two processes include identifying a business opportunity, developing a product or service, and founding a firm from the ground up.

Second, both technopreneurship and entrepreneurship require some level of inventiveness. Technopreneurs and entrepreneurs must be innovative thinkers to stand out in the market and offer value to their clientele. This could mean developing entirely new products or services or creating creative solutions to problems.

Thirdly, taking calculated risks is essential for both technopreneurship and commercial success. Technopreneurs and entrepreneurs alike must be willing to take calculated risks to launch and grow a business.

Technopreneurship and entrepreneurship share a set of necessary abilities: Technology-using entrepreneurs. In summary, there are a lot of similarities between technopreneurship and entrepreneurship. They both entail launching and expanding a company endeavour and call for a combination of leadership, business, and technological skills. The primary distinction between the two is that. In contrast, entrepreneurship can apply to various enterprises and industries; technopreneurship focuses on leveraging technology to create and innovate within the firm.

3.0.2 Importance of Technopreneurship

Technopreneurship, which enables business owners to introduce novel goods and services to the market and generate employment, is a major force behind economic growth and innovation. It's a developing industry with lots of potential for people with the know-how and drive to be successful technopreneurs.



Technopreneurship, which enables business owners to introduce novel goods and services to the market and generate employment, is a major force behind

Technopreneurship can resolve issues: Technopreneurs can create new products or services that meet unmet demands or challenges by applying their technical skills and knowledge to discover and solve problems.



Technopreneurship encourages innovation since these individuals are frequently at the forefront of developing technologies and can utilise them to develop cutting-edge goods and services. This has the potential to spur innovation, create opportunities for individuals to build capacity, and produce fresh, more effective approaches to solutions within our environment.

In-Text Question (ITQ)

Discuss the importance of technology infrastructure for technopreneurial ventures.

In-Text Answer (ITAs)

Technology infrastructure is essential for the success of technopreneurial ventures as it forms the foundation for their operations, scalability, and competitiveness. A robust technology infrastructure encompasses hardware, software, networks, and IT systems that support developing, deploying, and maintaining digital products and services. Technopreneurs rely on reliable and secure infrastructure to host their applications, store data, and deliver seamless experiences to users. Additionally, scalable infrastructure enables technopreneurs to adapt to changing demands, accommodate growth, and stay agile in dynamic market environments. Overall, a well-designed technology infrastructure is critical for the long-term sustainability and growth of technopreneurial ventures.

3.0.3 Technopreneurship Processes**Identify Business Opportunity**

There are numerous methods for spotting business possibilities, whether new ventures or entrepreneurial chances. Let's examine this:

Pay Attention To Any Changes In The Surroundings

A society is a dynamic organism that undergoes temporal and spatial changes. As a result, identifying business opportunities requires monitoring social, political, economic, and technological changes.



Techniques for Recognizing Commercial Opportunities

Nigeria offers many business options due to its broad market and population. However, how can you find the right business opportunity for you? Here are a few strategies for spotting business prospects.

Scanning the Environment

Monitoring outside influences on a firm, such as political, technological, and economic trends, is known as environmental scanning. Recognising potential dangers and possibilities that could impact the firm in the future is beneficial. Keeping a careful eye on how the environment is changing might help businesses grow or find new markets.

SWOT Evaluation

A tool for strategic planning, **SWOT** analysis assists companies in determining their advantages, disadvantages, opportunities, and threats. It can be used to create an action plan and assist in identifying internal and external issues that may impact the organisation.

Idea Generation

To continually discover new methods to add value for clients, innovation entails investigating novel technology, procedures, or business models. Combining a group of individuals to brainstorm can help generate new ideas and possibilities for goods, services, or business models. This is known as innovative brainstorming. It can assist in finding new avenues for growth and expansion, such as potential internet ventures for your present company or the automation of some processes, such as inventory planning.

Market Analysis

To find possible areas for growth or undiscovered markets, market research entails examining data on customer preferences, market trends, and industry statistics. You can find chances for an entrepreneurial business to provide your clients with the goods or services they want by conducting market research.

Marketing And Sales Strategies



What is a marketing and sales strategy?

A business sales and marketing plan is implemented to expand the business and attract new clients. While the marketing strategy concentrates on market research and brand development, the sales strategy is used to sell the products and is primarily concerned with reaching the client.

What Goes Into Developing a Sales And Marketing Plan?

There are several ways to develop a sales and marketing plan, but most strategies involve a few steps. The target audience must be determined, an analysis must be done, goals must be made, **KPIs** must be understood, budgets must be identified, strategies must be put into action, and content must be created.

Observing & Hearing Social Media

Social media and other online platforms are observed or monitored as part of social listening and monitoring to monitor competitor activity, consumer preferences, and market trends. It can help identify the best business opportunities in Nigeria for improving products and services.

Formulating a marketing plan known as the four Ps, the components of the marketing mix are used in a marketing strategy.

The components of the marketing mix are:

- Product - The product element concentrates on the product, its attributes, and how it benefits the user.
- Price: The price of a thing is determined by its cost and value.
- Place: Place is concerned with the product's route to reach customers.
- Promotion: All selling and advertising methods used to advance a product are collectively called promotion.

Funding and Financing:

These are two different ideas in the context of infrastructure investment. The act of acquiring or providing funds or capital



for a purchase or business venture is known as financing. Money given, especially by a government or institution, for a specific purpose is referred to as funding.

Creating And Managing a Team

You must be acutely conscious of your leadership style and methods as the team's leader. Do they work as well as you believe? How well-liked are they by the group you're trying to manage? Assess your performance and be critical of your improvement areas, particularly those that will help the people you manage.

Get To Know Your Team

Knowing your team inside and out entails learning about their cognitive processes and what it takes to inspire them to perform above and beyond expectations.

Establish Clear Roles And Obligations

Every team member must clearly understand their obligations and how they relate to one another. This is comparable to team sports, where certain players are called "system players"; they may not be the most skilled on the team but are adept at operating inside the "system." Because of this, you need to be able to see talent quickly and assess candidates not just on their suitability for a given function but also on whether they will work well in a team and fit into the workplace culture.

Take the initiative When Giving Feedback.

Any team needs feedback to ensure it remains on course and, more crucially, it is improving daily. Feedback needs to be ongoing and proactive. Many leaders tend to put off giving feedback until an issue arises.

Reward and Acknowledgment

People enjoy receiving praise, but they value respect the most. Give your teammates the recognition and respect they merit by taking the time to do so.

Legal And Ethical Considerations

An action that complies with the law is considered legal, but morality deals with proper and improper behaviour. This



implies that certain behaviours may be lawful but unethical in the eyes of others.

The following are important moral and legal factors for community service providers:

- Avoiding harm
- Assisting vulnerable populations
- Assent
- Confidentiality and privacy
- Particular legal obligations

Project Management At The End

- Project Start-Up
- Project Timetable Development
- Monitoring, Controlling, and Evaluating Projects

3.0.4 Technopreneurship Examples

Software Engineering

A technopreneur might start a software engineering company and build specialised software for enterprises or software products sold to a wide spectrum of consumers.

Development Of Mobile Apps

A technopreneur might, for instance, develop an online banking program that satisfies customer demand or a ride-sharing app like Uber or Boat. Internet-based marketplace: A technopreneur might establish an online marketplace that links vendors and consumers of a certain good or service. These marketplaces include sites like Konga and the Kusnap app store, where users can purchase and sell handcrafted goods and plan trips.

Online Store

A tech entrepreneur may launch an online store that offers a particular product or range of products, such as apparel or home goods. These are just a few examples of technopreneurship, and there are many other ways that technology can be used to start and grow a business.

3.0.5 An Illustration of A Prosperous Technopreneur



TECHNOPRENEURS

- Elon Musk
- Mark Zuckerberg
- Jeff Bezos
- Sergey Brin & Larry Page
- Steve Jobs
- Satya Nadella

Elon Musk, **CEO** of SpaceX and Tesla, is one successful technopreneur. After co-founding the online payment company PayPal, Musk launched several lucrative tech companies, such as SpaceX, which develops and builds cutting-edge rockets and spacecraft, and Tesla, which makes electric cars and renewable energy goods.

Elon Musk is renowned for his creative approach to business and his aptitude for problem-solving. He has constantly shown that he can introduce innovative products to the market, upending established sectors. He is a well-known and prosperous technopreneur due to his enterprises' notable development and disruption of traditional business methods.

Virtual reality: A technopreneur may find a business that develops immersive virtual reality games or training programs.

Mark Zuckerberg is the **CEO** and co-founder of Facebook, the biggest social networking site in the world. He founded the business when he was a Harvard University student, and since then, he has built it into a massive worldwide enterprise with billions of users.

Jeff Bezos is the **CEO** and creator of Amazon, one of the biggest online merchants in the world. In 1994, He started the company as an online bookstore, but it has now expanded to sell various goods, including clothes, gadgets, and home goods.

Sergey Brin and Larry Page: Google, the biggest search engine in the world; during their time as Stanford University students, they founded the business, which they have since expanded into a multinational technology conglomerate offering a vast array of services and goods, like as cloud computing, email, and maps.

Steve Jobs: One of the most inventive and successful technological firms in the world, Jobs served as **CEO** and co-founder of Apple. He was well-known for his ability to introduce cutting-edge items to the market and is recognised



for contributing to the computer, music, and smartphone revolutions.

Satya Nadella is the **CEO** of Microsoft, one of the world's biggest and most prosperous technological firms. He has guided the business through rapid expansion, concentrating on enhancing its cloud computing and artificial intelligence capacities.

In-Text Question (ITQ)

How do technopreneurs leverage technology to gain a competitive advantage?

In-Text Answer (ITA)

Technopreneurs leverage technology to differentiate their offerings, improve operational efficiency, reach global markets, gather and analyse data for informed decision-making, automate repetitive tasks, and continuously innovate to stay ahead of competitors.

3.0.6 A Successful Technopreneur's Attributes

The following characteristics are frequently linked to prosperous technopreneurs:

- **Technical proficiency:** Technopreneurs are highly knowledgeable about technology and can apply it to solve issues and add value.
- **Technopreneurs possess a solid grasp of business concepts,** such as marketing, sales, finance, and operations. This is known as their business acumen.
- **Technopreneurs possess innovative thinking skills,** enabling them to recognise and resolve issues and develop novel and inventive products or services through technology.
- **Risk-taking:** Technopreneurs are willing to take calculated risks to succeed since they know that starting and growing a firm involves some risk.
- **Persistence:** Technopreneurs can persevere through difficulties and disappointments. Adaptability:



- Technopreneurs possess adaptability and may quickly pivot or alter direction in response to shifting client demands or market conditions.
- Leadership: Technopreneurs are capable of inspiring and motivating their teams as well as directing the expansion and course of the company.
- Robust work ethic: Technopreneurs are devoted, industrious, and prepared to invest the time and energy required to succeed.
- Excellent communication skills: Technopreneurs can establish and preserve relationships with partners, employees, and consumers by successfully communicating their ideas and vision to others.

To put it briefly, successful technopreneurs are proficient in their fields and possess a solid grasp of business principles. They are also creative, tenacious, flexible, and capable leaders with excellent communication abilities.

3.1 Self-Assessment Questions

1. What challenges do technopreneurs face in harnessing technology for their ventures?
2. How can governments and institutions support the development of technopreneurship through technology?

3.2 Conclusion

Technopreneurship represents the dynamic fusion of technology and entrepreneurship, embodying innovation, agility, and a forward-thinking mindset. It catalyses economic growth, societal advancement, and transformative change. As technology continues to evolve rapidly, the opportunities for technopreneurs are boundless, offering the potential to revolutionise industries, create impactful solutions, and shape the future of our world. Embracing technopreneurship requires a combination of vision, adaptability, and a willingness to challenge the status quo, ultimately driving progress and prosperity in the digital age.

3.3 Summary



Technology serves as a catalyst for the development of technopreneurship by fostering innovation, providing access to resources, enabling scalability, facilitating collaboration, improving efficiency, and driving market disruption. As technology advances, it will play an increasingly vital role in shaping the entrepreneurship landscape and driving economic growth.

3.4 Unit Activities

1. **Team Formation:** Form teams of 3-4 members each.
2. Topic **Selection:** Each team selects a specific area within the intersection of entrepreneurship, social media, and **cybersecurity**. Examples include:
 - Developing a **cybersecurity**-focused social media marketing strategy for a startup.
 - Creating a social media platform with enhanced **cybersecurity** features.
 - Analysing **cybersecurity** risks and opportunities for established social media companies.
 - Exploring the role of social media in **cybersecurity** awareness and education. (Complete the activities within a week)

3.5 Tutor-Marked Assignment

Define technopreneurship and discuss its significance in the modern business landscape.

Submit your responses to Dropbox for grading.

10.1 MODULE 2: SOCIAL MEDIA MARKETING

10.1.1 UNIT 2: SOCIAL MEDIA MARKETING

1.0 Introduction



Technology has transformed social media marketing, reshaping how companies engage with audiences across diverse platforms. Platforms like Facebook, Twitter, Instagram, and emerging contenders like X platform and TikTok have become essential for businesses. Entrepreneurs leverage



these platforms to directly connect with their audience, driving brand awareness and growth. Social media's engagement and analytics empower businesses to refine strategies with precision, shaping the future of commerce in the digital age.

2.0 Learning Outcomes

At the end of this unit, you should be able to:



- Understand the impact of technology on social media marketing strategies.
- Recognise the importance of real-time engagement across various digital platforms.
- Evaluate and apply effective social media tools for building brands and driving business growth.

3.0 The Roles Of Technology In Social Media Marketing

Innovative marketing campaigns and strategies have been made possible by technological developments in marketing, which have also completely changed how companies interact with their target audiences.

Marketing is changing as we spend more time on our laptops, tablets, and mobile devices. For companies, the challenge is to engage with consumers in real time across all of these platforms and develop effective campaigns for e-commerce, social media, and display advertising.

The nature of marketing has altered due to the real-time conversations firms have with consumers through mobile apps and websites.

3.0.1 Social Media: What Is It?

Various technology tools that enable individuals to exchange ideas and information are called "social media."

Twitter (X platform), YouTube, Facebook, Instagram, and approximately 60% of the world's population, or over 4.7 billion people, use social media.

3.1 Social Media's Development and History



MySpace reached one million users monthly first, in 2004. Tom Anderson founded MySpace 21 years prior to Facebook and Instagram.

Initially intended to communicate between friends and family, social media quickly grew to serve many functions. Myspace was the first network to hit one million active monthly users in 2004 (Ortiz-Ospina, 2023).

Myspace existed before Facebook and Instagram revolutionised the internet. Tom Anderson created the social networking site 21 years ago.

Users from that era may recall that it served as a platform for budding indie bands to share their music with the world and to interact with friends online. Bands like My Chemical Romance, Paramore, and Arctic Monkeys all had early accounts.

Social media use skyrocketed after the release of Facebook and Twitter (now known as the X platform) in the next years. Businesses rushed to these platforms to quickly connect with a global audience.

With Twitter rebranded as "X," Elon Musk envisioned it as the next "everything app." In 2022, Musk paid \$44 billion to purchase Twitter. The iconic blue bird logo was replaced with a stylised "X" mark on July 23 when Twitter removed the Twitter name from all of its websites, apps, and corporate offices. Redirecting even X.com to Twitter.com is the new setup.

Social media is where 46% of internet users acquire their news, according to the Global Web Index. Conversely, more than any other generation, Gen Z and Millennials were the most inclined to consume news on social media.

Considering how much time people spend on social media sites and applications daily, it is unsurprising that social media plays a big part in many businesses' marketing strategies. Simultaneously, social media is constantly evolving as relatively new applications such as Telegram and TikTok have entered the ranks of well-known social networks like Facebook, Instagram, YouTube, and X platform.

In-Text Question (ITQ)



How can technopreneurs use social media to build brand awareness?

In-Text Answer (ITA)

Technopreneurs can use social media to build brand awareness by consistently sharing relevant and engaging content, interacting with their audience through comments and messages, collaborating with influencers or industry experts, hosting contests or giveaways to encourage user participation, and utilising visual content such as videos and infographics to capture attention and communicate their brand's message effectively.

3.2 Use of Social Media for Entrepreneurial Marketing

Social media is a tool entrepreneurs use to market their goods and services. Social networking is a great way to advertise your goods or services. Before the invention of social media, it was impossible to develop a brand, boost sales, and interact with people in the same ways you can now. Entrepreneurs want to think about how they can provide engaging material to draw readers to take advantage of this new technology.

Social media offers many advantages for business owners who wish to see growth and success in their online ventures.

They can generate talk about their company by providing details about themselves or their goods and services on social media sites like Facebook, Twitter, and Instagram. Establishing a direct connection with present and potential customers who could be interested in purchasing from these businesses shortly enables them to reach a larger audience than ever!

3.2.1 What You Can Do with Social Media in the Modern Age

Social networking, social bookmarking, social news, online forums, and media sharing are the six primary types of social media platforms. Many different user goals and preferences are satisfied by these diverse platforms. Professionals shouldn't use some of these; amateurs should. Others use them to locate people worldwide with similar political or other views.



Social media enables firms to build client relationships, fostering trust, and provides a platform for customer feedback.

Social media acts as a bridge between funders and charity, celebrities and followers, and politicians and voters. During emergencies, social media is a common tool used by governments to disseminate important information.

3.3 Why Use Social Media for Business?

You may communicate with your customers using social media. Instead of waiting days or weeks for a customer support agent to get back to you, you may listen to what they have to say and answer instantly. Social media allows companies to establish relationships with their clients, which is crucial since it fosters mutual trust. Furthermore, social media gives customers a voice when dissatisfied with a good or service, allowing companies to grow from their mistakes and do better when interacting with customers in the future.

Business owners can grow their audience by using social media as a tool. Social media is a tool that entrepreneurs use to interact with customers, build their brand, get their opinions, and promote their products. As an entrepreneur, you know what it's like to be at the forefront of your business. You interact with consumers daily by phone, email, and in person. Through social media, you can get in touch with them in a different way and perhaps even meet new people. It helps reduce the time spent answering questions by creating a central location where people may ask them rather than waiting for them to come up one-on-one during events or meetings.

Social media's unmatched potential in three key marketing domains—connection, interaction, and customer data—powers social media marketing.

3.4 Self-Assessment Exercise

1. What role does data analytics play in social media marketing for technopreneurs?
2. How can technopreneurs leverage social media to drive sales and conversions

3.5 Conclusion



In conclusion, using social media for entrepreneurial marketing is no longer optional but essential for success in today's digital landscape. Entrepreneurs who effectively harness the power of social media can gain a competitive edge, establish their brand identity, and cultivate meaningful relationships with customers. However, entrepreneurs must understand their target audience, choose the right platforms, and consistently deliver high-quality content to maximise the benefits of social media marketing. With strategic planning and execution, social media can be a powerful tool for driving growth and achieving long-term success in entrepreneurship.

3.6 Summary

The use of social media for entrepreneurial marketing has become a vital strategy for businesses of all sizes. Social media platforms offer unparalleled reach, allowing entrepreneurs to connect with their target audience cost-effectively and efficiently. By leveraging platforms like Facebook, Instagram, Twitter, and LinkedIn, entrepreneurs can build brand awareness, engage with customers, and drive sales.

Social media enables entrepreneurs to create targeted advertising campaigns, tailor content to specific demographics, and track the effectiveness of their marketing efforts through analytics tools. Additionally, platforms like YouTube and TikTok provide opportunities for entrepreneurs to showcase their products or services through video content, reaching a wider audience and fostering brand loyalty.

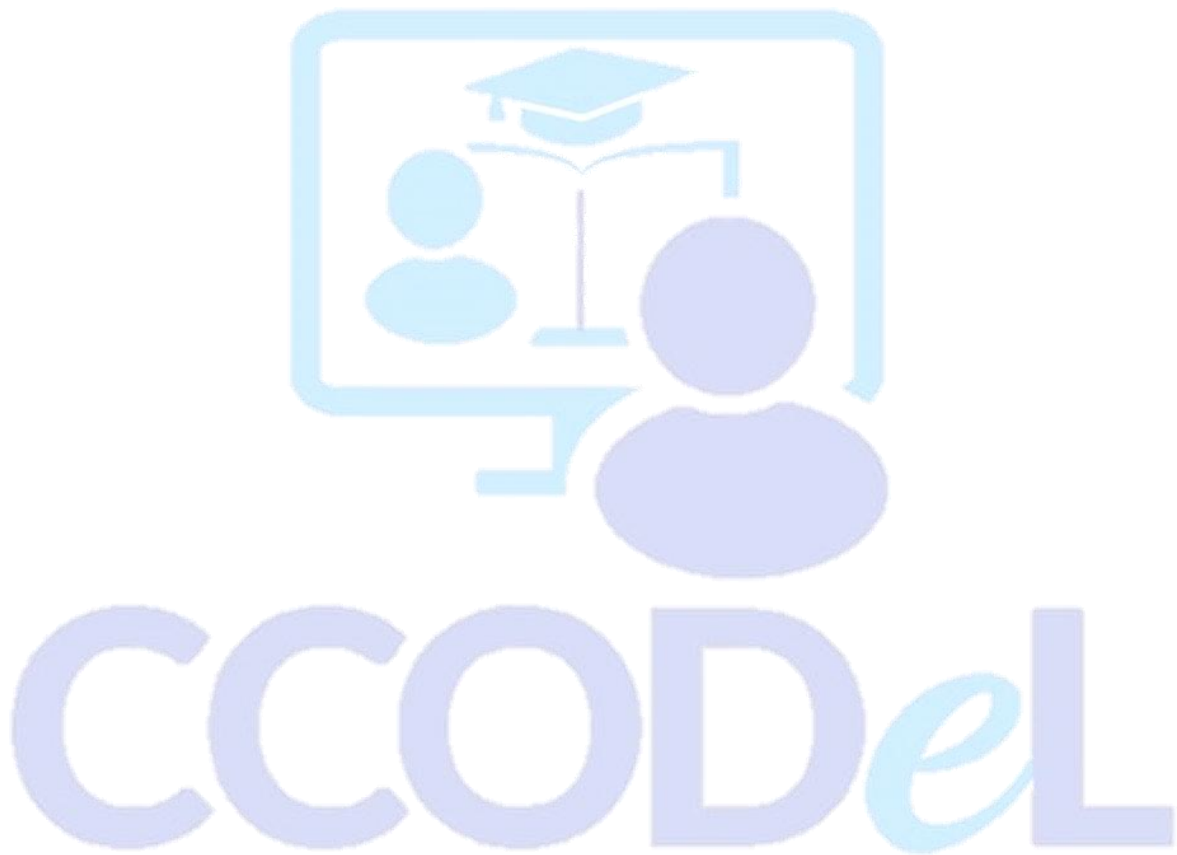
3.7 Unit Activities

Research and Analysis: Teams research their chosen topic, gathering insights from academic literature, industry reports, case studies, and relevant news articles. They should analyse the current landscape, identify challenges, and assess opportunities.

3.8 Tutor-Marked Assignment



Market Research: Teams research their chosen topic within the time frame.





10.2 MODULE 3: LEADING SOCIAL MEDIA NETWORKS FOR COMPANIES

10.2.1 UNIT 3: LEADING SOCIAL MEDIA NETWORKS FOR COMPANIES

1.0 Introduction



Social media is essential for engaging customers, driving sales, and building brands in the modern business landscape. With billions of users on platforms like Facebook and TikTok, businesses can reach vast audiences. Automation streamlines processes, freeing managers to focus on strategy. Staying agile is crucial in this dynamic environment, ensuring competitiveness in the digital age.

2.0 Learning Outcomes

At the end of this unit, you should be able to:

- 
- Understand the significance of social media in contemporary business operations
 - Develop strategies for effectively utilising various social media platforms for marketing and communication.

3.0 Social Media Is a Necessity

Social media has become a vital instrument for promoting businesses. Businesses use it to locate and communicate with clients, promote and advertise to increase revenue and spot rapidly evolving technology.

3.1 The World's Top 10 Social Media Sites

According to Data report, the World's top 10 Social Media Sites as of January 2023 include:

1. Facebook, with 2.96 billion users
2. YouTube, with 2.51 billion users
3. WhatsApp, with two billion users
4. Instagram, with two billion users



5. WeChat with 1.31 billion users
6. TikTok, with 1.05 billion users
7. Facebook Messenger, with 931 million users
8. Douyin, with 715 million users
9. Telegram with 700 million users
10. Snapchat, with 635 million users

3.1.1 Which social media platform is the most useful?

According to a recent estimate, Facebook is the most widely used social networking platform worldwide, with an estimated 3 billion users.

What is the average time spent on social media?

According to Data Reports, Internet users between the ages of 16 and 64 reportedly spend an average of two hours and thirty-one minutes per day on social media as of late 2022. Additionally, the survey states that "nearly four out of every ten minutes spent online can now be attributed to social media activities, accounting for social media's highest share of total online time to date.

3.1.2 Social Media Marketing: What Is It?

Use of social media for entrepreneurial marketing.

Social media marketing refers to using social networks for sales and marketing. Establishing an online presence and engaging with customers immediately are two of the many ways that businesses can advertise their products or services.

Local businesses now have more options to connect with potential clients through social media marketing without shelling out a premium for ad placement.

For business owners, social networking is a highly helpful tool. It's a way to build your brand, build relationships with customers, get feedback from them, grow your community, and much more.



Social media marketing involves utilizing social networks to promote products/services by establishing an online presence and engaging customers.



There are claims that social media has replaced traditional marketing strategies such as print or TV ads as the main way companies reach out to their target audiences (and even people who aren't customers).

Social media marketing is more important than ever for business owners these days. These days, there is fierce rivalry in every field, and to stand out from the crowd, a company needs to market itself on social media.

SOCIAL MEDIA MARKETING PLATFORMS				
	PEOPLE	CONTENT	STRATEGIES	CONS
	• 25-34 • Boomers	• Photos & links • Information • Live video	• Local mktng • Advertising • Relationships	• Weak organic reach
	• 18-25 • 26-35	• How-tos • Webinars • Explainers	• Organic • SEO • Advertising	• Video is resource-heavy
	• 18-24, 25-34 • Millennials	• Inspiration & adventure • Questions/polls	• Ecommerce • Organic • Influencer	• High ad costs
	• 25-34, 35-49 • Educated/wealthy	• News • Discussion • Humor	• Customer service • Ads for males	• Small ad audience
	• 46-55 • Professionals	• Long-form content • Core values	• B2B • Organic • International	• Ad reporting & custom audience
	• 10-19 • Female (60%)	• Entertainment • Humor • Challenges	• Influencer marketing • Series content	• Relationship building
	• 13-17, 25-34 • Teens	• Silly • Feel-good • Trends	• Video ads • Location-based mktng • App mktng	• Relationship building
				 WordStream by LOCALIQ

The foundational element of every Digital Marketing endeavour is implementing an appropriate plan. To build a powerful social media marketing campaign, businesses using social media marketing must ideally integrate all the significant elements of their web domain.



It is a well-known truth that when people are online, they most frequently explore social media platforms. Most of the time is spent reading blogs and websites for information, watching videos, and listening to music podcasts.

3.2 How to Utilize Social Media at Work

As a working professional, we understand that you don't have enough time to visit every useful site on our list.

A comprehensive summary of all the ways social media could help your business can be seen below:

Increase brand awareness

Companies can create and share social media material highlighting their brand. Reports about your amazing deeds may begin to appear online if you regularly post pertinent and engaging content.

Interact with your clients directly.

Does social media provide businesses with any other benefits? Through social media, companies and their customers can communicate directly. Businesses may interact more directly and personally with their audience on Facebook, Instagram, and X than they can via email or a hotline.

Advertise your work

As your primary source for publishing, think of social media. Submit a blog entry on your **X** account, utilise Facebook to drive readers to your blog entries, and submit a video on TikTok. Use social media to your advantage to get readers and potential customers to see the entertaining or instructional content you've spent a lot of time and energy creating. Generate new sales leads.

Potential clients may be discovered on social media daily. Relationship-building and social marketing are made possible by social media networks., together with the opportunity to target and market. To create a fantastic database of sales leads, concentrate on targeting audiences according to their demographics, interests, and behaviours.

**Analyse the market**

You'll have access to a wealth of amazing market intelligence if you become proficient with social media analytics tools. Collaborating with influential people and disclosing consumer preference data can be achieved through social media.

Over the past ten or more years, social media sites have established their celebrity universe. They are informed, influential people with responsive audiences. Wise businesses understand just how to harness their potential. If your business uses social media, consider joining forces to launch a campaign. Or a long-term impact ambassador strategy. Our guide on influencer marketing is an excellent resource for learning about this exciting new industry.

Be mindful of the image you have

Even if you haven't spent every moment on social media, your clients and supporters have likely, as well as potential detractors. Uninformed people may spread rumours about you on X, slagging your product on Instagram Stories, or negatively talking about you. Who knows what might happen if you're not around to control the conversation?

On Lululemon's Instagram postings, as shown below, comments are left by both ardent supporters and detractors; however, the social media team controls opinions as well as actions to assist you in making informed decisions about your products and services both in person and online, as shown below.



Take note of your competitors.

See what your competitors are doing and what doesn't work. Get a perceptive understanding of your intended audience. Social media monitoring, social streaming, and competition analysis are used to spot market trends and innovative strategies.

Show off your ROI

It's always difficult for marketing experts to show a profit on investments. But people can see the full impact of all you do on social media, with the aid of social media research and monitoring tools, from fans to engagements to transactions.

Google Analytics and Hoot Suite Impact monitor internet traffic using social media, conversions, email sign-ups and return on investment (**ROI**) for organic and funded online marketing.

YouTube is the world's second most popular website. Additionally, even though we now view it as a means of pleasure, it is a platform for socialising in technological concepts. Social media platforms allow everyone to connect



Google Analytics and Hootsuite Impact are tools that track internet traffic, conversions, email sign-ups, and ROI for online marketing via social media.



with others, engage with producers, post material, and keep friends and preferred businesses updated.

YouTube offers some intriguing advantages for businesses, just like all the other well-known players in the social media space.

People from the community can congregate on your web page. Connect with your viewers by using YouTube's live-streaming feature.

You might be able to make money off of your work by having other people advertise it if it gets views. Using Google AdSense, YouTubers often earn \$3 to \$5 for every 1,000 views on their videos.

In-Text Question (ITQ)

How can social media be used for communication at work?

In-Text Answer (ITA)

Social media platforms can serve as channels for internal communication among employees, allowing teams to collaborate, share updates, and coordinate tasks in real-time.

3.3 Using Facebook, Instagram, And Chat Apps for Business Purposes



Meta owns Messenger, WhatsApp, Instagram, and Facebook, leading to extensive user engagement across these platforms, with 3.65 billion monthly users.

Meta is the owner of Messenger, WhatsApp, Instagram, and Facebook. As a result, there is a large user base and a lot of interaction between these platforms. Data indicates that 3.65 billion people use at least one Meta service monthly.

One great social media platform to market to is Facebook.

- baby boomers
- holding discussions and live broadcasts
- announcing hours' changes, events, and milestones
- cultivating relationships with current consumers.

Since Facebook's organic reach is constrained, Facebook advertising is your best option if you're trying to discover new audiences or generate leads.

3.3.1 Tiktok For Company



- TikTok is the most popular platform for marketers on the internet, with over one billion monthly active users. There are several ways to engage with the community naturally on the short-form video platform through challenges and trends. However, a little goes a long way if you have a few bucks.
- To reach 884.9 million viewers, employ TikTok adverts. You can choose between in-feed and video shopping adverts
- The younger generation now frequently uses TikTok as a search engine. Therefore, companies may gain a lot by joining this network and improving their social media optimisation techniques.
- Share your content to have a wider reach.
- Marketers may establish a strong connection with creatives and users on TikTok who are already thriving on the platform by using the Developer Network.

3.3.2 X For Business Purposes

- **X**, originally referred to as Twitter, is a platform for messaging via text that connects people and organisations, customers, and ambassadors for the company. **X** internal data indicates that 76 per cent of participants agree that discussions on the platform have motivated them to buy goods.
- You may offer committed customer assistance on X by sending quick DMs to customers.
- X Lists are an excellent tool for keeping tabs on your sector or competitors.
- Whether you want to increase brand awareness, boost traffic to your website, or create leads, advertising on X can help you reach a wider audience and engage with them more deeply.

3.3.3 Pinterest For Companies

Even though Pinterest isn't as well-known as TikTok, it may still be a fantastic social media tool for companies. If, of course, the audience—all 465 million—fits your brand.



According to Pinterest Company, internet marketers utilise the website's features to:

- Enlarge their internet footprint and attract new customers.
- Boost the number of people visiting their company's homepage or digital store.
- Encourage engagements like purchases, reservations, and website sign-ups.

3.3.4 For Business, Snapchat

Do you want to establish a connection with the children? The majority of the 319 million daily active Snapchat users are under the age of 35.

Among the distinctive features of Snapchat for brands are:

- Producing a branded geo-filter
- Developing a sponsored augmented reality lens
- Promoting on Snapchat using a range of ad formats, such as "Swipe Up to Call" and shop ads.

3.3.5 The Search Engine Corporate Profiles For Businesses

You should participate in the discussion here because Google is one of the unique social media platforms designed specifically for businesses.

Why does Google Business Profile have such power?

Maintaining an updated profile is a tiny time commitment that yields significant rewards:

- More consumers
- Consumers can rapidly get an impression of your brand from your images.
- Evaluations and improvements.
- Consumers looking for a company like yours come across your web page.

3.4 Business Suggestions for Social Media

Here's how to use the social media accounts for your business to its full potential:

Make a clear social media objective.



Measuring success and return on investment (ROI) is nearly impossible without clearly stated goals.

Before you post often, ensure your social media marketing goal meets these requirements: it must be clear, tangible, achievable, applicable, and time-bound.

It's critical to resist getting sucked into vanity metrics, such as the number of likes and follows, as these figures might not accurately represent the worth of your work. Rather, pay attention to important data such as conversion, click-through, and engagement.

Do you require some instances of SMART social media goals? We have you in our hands.

Recognise your target audience.

You can learn a lot of useful information about your following from social media analytics. Examine the data to gain insight into interactions with your brand, regional distribution, and demographics.

Maintain coherence

Users on social media adore a reliable person. If you consistently deliver material, you will reap the rewards. Sharing material frequently puts your business in the forefront of your followers' minds and helps to establish brand awareness. Sustain a regular blogging schedule to keep readers interested.

Provide engaging content for readers.

You could update Facebook with a hundred pointless posts every day. Alternatively, you may concentrate on producing worthwhile, excellent content that genuinely resolves problems, makes people laugh, teaches, or inspires. We know the one we want to see more of in our feeds.

Attempt to establish a genuine connection rather than sharing excessively promotional content. You have a much better chance of leaving a lasting impression and gaining a new audience member if you make them smile, educate



them, or present them with a Choose the appropriate content combination.

Ensure the mission statement you've allocated to each social media page is reflected in your content strategy and calendar so that everything you post supports your business objectives.

You may determine that:

50% of the material will increase website visitors.

25% of material curation will be from outside sources.

Lead-generation objectives (such as newsletter sign-ups and eBook downloads) will be supported by 20% of the content.

Your company culture will be covered in 5% of the content.

By planning these different post types in the content schedule at the appropriate times, you can ensure that you maintain the right balance.

An economic principle states that, for any particular occurrence, 80% of the results, or final products, arise from 20% of the factors that led to it or sources.

Your posts must include information, entertainment, or education for eighty per cent of the time. Twenty per cent, however, can immediately benefit your business.

Following visual media

Although we read a lot to be informed, humans are primarily visual. Give your audience something eye-catching. Posts with eye-catching graphics, photos, and videos can entice visitors to interact with your content while they scroll down.

Engage with your supporters.

Instead of using social media as a broadcasting platform, use it as a two-way conversational exchange. Respond to mentions, messages, and comments as quickly as you can. Encourage discussion and feedback on your work. Developing a rapport with your audience fosters trust and confidence.

**Develop keywords and strategies for search engine optimisation on social media.**

Even if search functions on many platforms improve, tagging remains useful for increasing visibility and drawing in more people.

To make certain to use no more than three to five keywords per caption, research the most common ones relevant to your content and your audience's interests. Perfecting your discoverability techniques is necessary to grow your customer base.

Arrange contests and gifts. The reality is that Free things are popular

Plan imaginative and captivating gifts or competitions to increase engagement and attract new fans. It's a great method to gain additional visibility and encourage your current audience.

Make use of your data analysis information. Authority comes from information!

Use social networking tools to track the success of your postings or campaigns. You and your team can determine what functions best by analysing your successes and mistakes. After that, you can adjust your strategy to see those crucial indicators gradually increase.

Post with personality and sincerity.

Nobody visits social media sites to watch dull commercials. Become authentic and imaginative. In all your interactions and discussions, be sincere. Customers value open and honest firms, and transparency fosters confidence.

Share material created by users.

We understand it can be intimidating to stare at a blank social media dashboard for a while. However, you won't ever have to worry about writer's block again if you receive a consistent flow of user-generated material. Sharing user-generated material can serve as social proof in addition to keeping your content calendar nice and full.

**Assemble a coalition of influences.**

Lean on the strength of those who excel in social interactions. To greatly increase your reach and reputation, collaborate with influencers in your niche who share the same values as your company.

Engage in social listening.

A little listening goes a long way when using social media for business. Create streams to monitor conversations about your company, goods, and sector. The monitor said good and bad things! Social listening enables you to effectively read client sentiment and reply to criticism like a superpower.

Try out some live-streaming

More genuine and immediate interaction with your audience may be achieved using live videos on social media sites like Facebook, Instagram, and **X**. They can be used for anything, such as giveaways and marketing surveys or questionnaires.

Observe the competitors

The proverb "Keep your enemies and your friends close" is well-known. This is especially true with social media. Astute companies monitor their rivals' social media activity to take note of their triumphs and errors.

The online world is a constantly changing environment. You must stay current with trends if you want to remain relevant. Watch the Explore page, pay attention to online tastemakers, and participate actively.

It's also critical to keep up with the constantly evolving algorithms of these social media sites; actions that provide rewards one day may yield completely useless results the next. To stay current on the most recent algorithmic developments, follow the Hootsuite blog and modify your social media strategy accordingly. The most recent tools, techniques, and insights for **X**, Facebook, Instagram, and other platforms may be found on Hootsuite's social media marketing blog.

3.5 Social Media Automation: What Is It?



Tools that automate post-scheduling, basic customer service, and analytics reports can save social media managers valuable time for higher-priority tasks. Utilising tools or software to handle repetitive chores on social networking websites, such as publishing content or creating reports, without requiring human intervention is known as social media automation. You may always use generative **AI** tools to develop content ideas if needed.

3.5.1 Social media scheduling and publication

Changing between social media accounts to post several times a day can be very time-consuming. Particularly since platform-specific best practices for content and the optimal time to post differ.

Posting on social media automatically boosts productivity and saves time without sacrificing the quality of the information. You can utilise automated social media posting to plan the right pieces and post times on each network after your content is protected and loaded. After that, just unwind and watch the robots move your posts across the finish line. [Go to this URL](#) (for more information).

3.6 Self-Assessment Exercise

1. What are some ways to use social media for marketing in the workplace?
2. How can social media help with networking in a professional setting?

3.7 Conclusion

Social media has revolutionised how individuals and businesses connect, communicate, and market themselves. By incorporating social media into their work strategies, professionals can enhance their brand, reach a wider audience, and achieve their professional goals more effectively. However, success on social media requires careful planning, consistent effort, and a deep understanding of your target audience. By following best practices, monitoring performance, and adapting to changes in the social media



landscape, individuals can harness the power of social media to drive their careers and businesses forward.

3.8 Summary

Utilising social media for work purposes has become increasingly prevalent in modern professional environments. By leveraging various platforms strategically, individuals and businesses can enhance their visibility, engage with their target audience, and ultimately achieve their goals more effectively. Here are key points to consider when using social media for work:

1. **Identify Goals:** Determine clear objectives for your social media presence, whether it's increasing brand awareness, driving website traffic, generating leads, or fostering customer relationships.
2. **Know Your Audience:** Understand your target audience's demographics, interests, and behaviours to tailor your content and engagement strategies accordingly.
3. **Choose the Right Platforms:** Select social media platforms that align with your objectives and where your target audience is most active. Each platform has strengths and user demographics, so prioritise those that best suit your needs.
4. **Create Compelling Content:** Develop engaging and relevant content that resonates with your audience. This can include posts, articles, videos, infographics, and more. Consistency and quality are key to maintaining audience interest.
5. **Engage and Interact:** Foster meaningful interactions with your audience by promptly responding to comments, messages, and mentions. Actively participate in conversations about your industry or niche to establish thought leadership.



6. **Utilize Analytics:** Monitor and analyse your social media performance using built-in analytics tools or third-party platforms. Use these insights to refine your strategies and optimise your content for better results.

7. **Stay Authentic and Transparent:** Maintain authenticity in your communication and interactions on social media. Be transparent about your brand values, offerings, and sponsored content or partnerships.

8. **Stay Updated:** Keep abreast of the latest trends, features, and best practices in social media marketing to adapt and stay ahead of the competition.

3.9 Unit Activities

1. Read the relevant content in the study guide. (30 minutes)
2. Consult the recommended texts and the World Wide Web for additional material to enrich your understanding. (30 minutes)
3. Complete the tutor-marked assignment below

4.0 Tutor-Marked Assignment

Proposal Development: Based on team research, you should teams develop a comprehensive proposal on the assigned topic.



10.3 MODULE 4: BEST SOCIAL NETWORKING SITES FOR COMPANIES AND PLATFORMS

10.3.1 UNIT 4: BEST SOCIAL NETWORKING SITES FOR COMPANIES AND PLATFORMS

1.0 Introduction



Social media management is vital for businesses today. Tools like Hootsuite Composer simplify tasks across major platforms like Instagram, Twitter, LinkedIn, and YouTube. However, challenges persist, from maintaining engagement to navigating platform algorithms. This unit'll explore social media marketing's nuances, pros, cons, and effective strategies for technopreneurship ventures. Let's uncover its potential for business growth.

2.0 Learning Outcomes

At the end of this unit, you should be able to:



- Understand the importance of social media marketing in technopreneurship ventures and develop effective strategies for leveraging social media platforms to achieve business objectives.

3.0 Social Networking Sites and Platforms for Companies

3.0.1 Hootsuite



Another invaluable resource for businesses is the Hootsuite Inbox, consolidating social media messages from various platforms into one interface, simplifying customer support and communication.

Organising, scheduling, optimising, and monitoring social media postings are just a few of the tasks that Hootsuite Composer's comprehensive platform manages. It also makes preparing and posting material on Instagram, Twitter, LinkedIn, YouTube, and other channels simple.

With Hootsuite, you can quickly reuse your top-performing articles, find and fix typos, add branded images, and get insightful suggestions for the best hashtags to use and when to post.

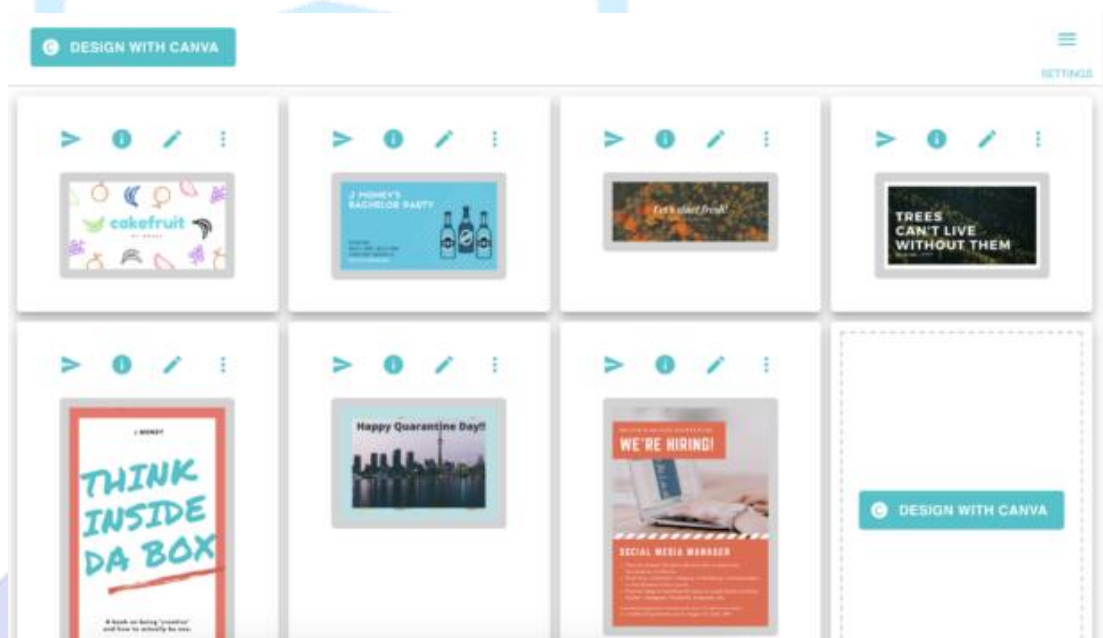
One more incredibly useful tool for businesses is the Hootsuite Inbox. Regardless of the platform, Inbox unifies all social



media messages and conversations into a single interface. Hootsuite Inbox makes customer support and discussions easier. It lets you quickly respond to inquiries and track a person's past interactions with your company, so you never have to leave a client hanging.

3.0.2 Canva

For creating distinctive and branded visuals, Canva is the ideal choice if you work alone or with a small social team and do not have access to a graphic designer. There's a good reason why social media marketers embrace it.

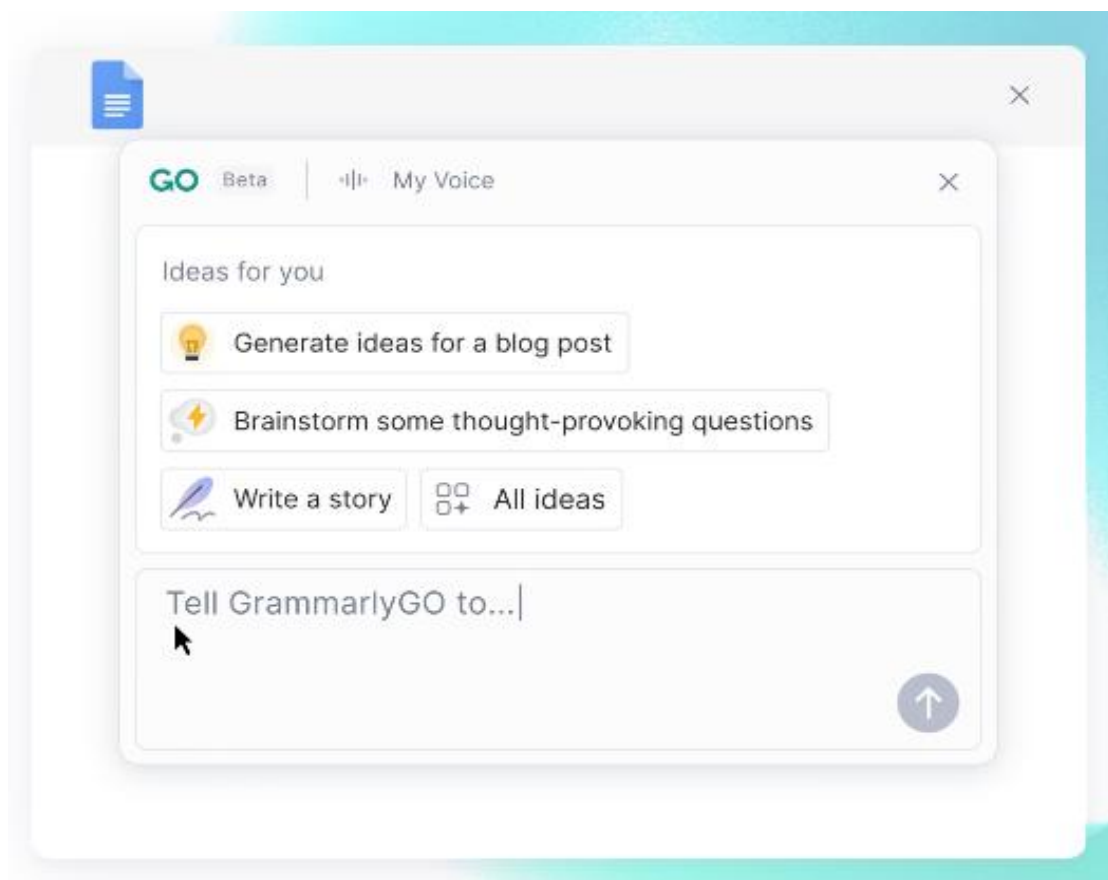


With Canvas templates and easy-clicking and dropping design features, someone with no experience in graphics can produce beautiful images and videos. [Go to this URL](#)

You may quickly upload using the assets you created directly in Composer once you subscribe to a Hootsuite account.

3.0.3 Grammarly

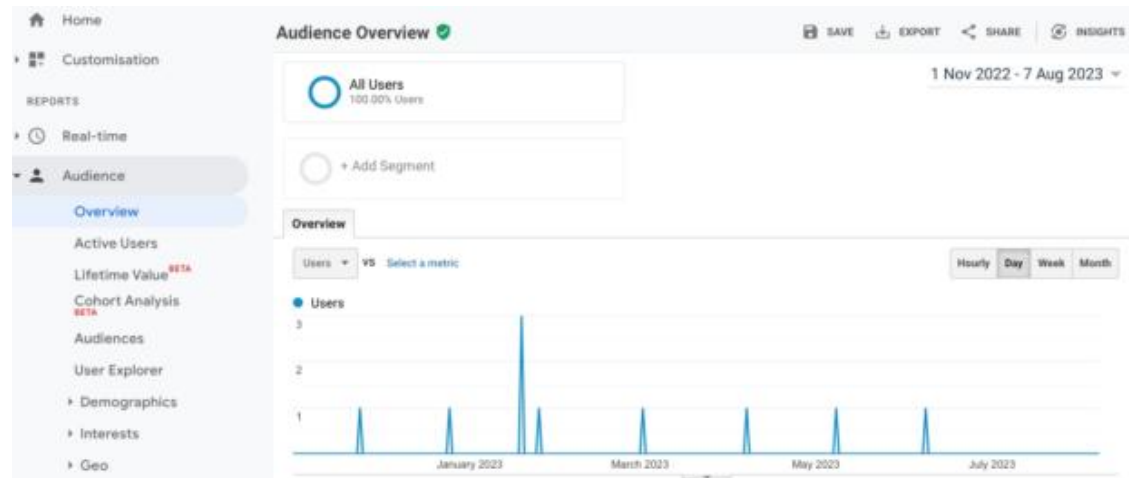
Mistakes in posts can make them appear careless and unclear. Grammarly can assist with proofreading, as shown in the figure:



In addition to helping with spelling and grammar correction, the **AI** technology can also assess material for clarity and tone. Grammarly has even added text-generation and brainstorming functions in their most recent upgrades. [Go to this URL](#)

3.0.4 Google Data

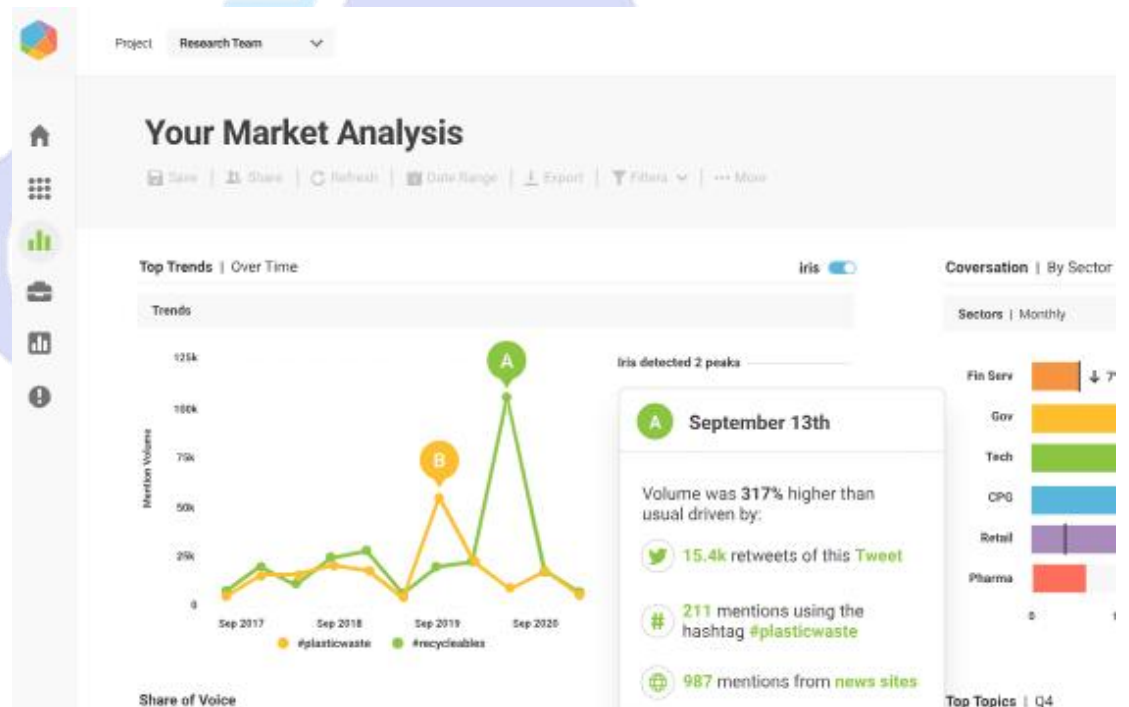
Google Analytics is initially free to use, but it is also essential. Google Analytics will be quite helpful for monitoring and assessing your company's traffic goals. Would you like to showcase how your social media and internet presence has been effective? Analytics is available to help.



Google Analytics gathers data on various issues, including demographics, traffic sources, leads that convert, and visits to certain pages. One of the most often used tools for digital marketers is Google Analytics.

3.0.5 Brandwatch

If your company is large, you should use Brandwatch's capabilities. When gathering market data and measuring brand emotion, Brandwatch is just outstanding.





The unified platform provided by the enterprise-level solution gives marketers access to priceless industry data. You can easily track brand performance measures and dive into customer insights with Brandwatch.

One of its best features is that it integrates seamlessly with Hootsuite, a social monitoring platform for businesses that is powered by Brandwatch and known as Hootsuite Insights.

Use Hootsuite to manage your social media presence and save time. You can locate relevant conversions, schedule content, interact with the audience, track outcomes, and more from a single dashboard.

3.1 Benefits and Drawbacks of Social Media Marketing (SMM) Benefits

With the emergence of online advertisement, there are now additional benefits. These online platforms provide an effective way to engage and communicate with many people, which helps to increase company visibility and recognition.

Digital connections with customers can improve viewer interactions and promote brand loyalty. This is more appealing to smaller or newly established businesses because it is frequently less expensive than conventional promotional techniques.

Additionally, there are several advantages to social media promotion, including links to your networking site profile. It can aid in raising exchange rates and increasing website traffic. Furthermore, social media offers a real-time platform for consumer opinion collection, facilitating quick engagement and conversation.

A further advantage of online marketing is its precise and broad ability. Admires, feedback, shares, and other forms of social media interaction may assist businesses to increase participation and attract a wider audience. This is particularly important because numerous individuals exchange information among close companions who could be different



Digital connections with customers enhance interactions and foster brand loyalty, especially for small businesses due to cost-effectiveness. Social media facilitates real-time feedback and targeted marketing.



viewers. On the contrary, social networking sites offer organisations a range of marketing choices that let them provide audiences with information specifically catered to their interests, practices, and backgrounds.

In-Text Question (ITQ)

How can businesses mitigate the risks associated with social media marketing?

In-Text Answer (ITA)

Businesses can mitigate risks by actively monitoring and responding to feedback, staying updated on platform changes, diversifying marketing strategies, investing in social media management tools, and focusing on providing genuine value to their audience.

3.2 Negative Aspects

Online marketing has not been without difficulties and issues despite its numerous advantages. Building a successful online presence takes a lot of work; entrepreneurs must publish material and interact frequently.

A successful social media advertising campaign requires a deep understanding of every marketplace available and the ability to create captivating material, analyse data, and make intelligent conclusions using the results. Since each system is unique, it often must be understood separately. Also, predicting and maintaining achievement might be difficult because social media firms regularly change their rules and procedures.

Social media facilitates better user engagement and gives people a public forum to express their concerns and thoughts. This may unknowingly spark an open conversation that, if handled poorly, may jeopardise the business's credibility.

Organisations lacking a lot of influence or recognition are trapped, pulling in ineffective users and producing barely any prospects due to how much they rely on social networks to market their business.



Lastly, it might prove difficult to calculate the exact value you'll get from digital marketing. Since online advertisement usually entails monitoring several indicators, analysing huge, complex data sets, and making predictions about possible consumer conduct, determining this approach's effectiveness and profit on investment can be difficult.

3.2.1 Pros

- It may make it simple for businesses to increase brand recognition and provide them with more affordable options and excellent visibility.
- Possibly used to boost real-time feedback and website traffic
- Able to be used for focused or particular interactions

3.2.2 Cons

- It could take a while to set up and maintain.
- It could be erratic because different platforms might have different algorithms
- , which can lead to unfavourable comments being made public.
- It could be challenging to completely comprehend the ROI.

3.3 Self Assessment

1. How can social media marketing benefit my business?
2. What are the drawbacks of social media marketing?

3.4 Summary

Social media marketing plays a pivotal role in the success of technopreneurship ventures, offering a dynamic platform for engagement, brand building, and customer acquisition. In the rapidly evolving landscape of entrepreneurship, leveraging social media effectively can lead to significant competitive advantages. Here's a concise summary of the key points:



1. **Amplified Reach:** Social media platforms provide technopreneurs with unparalleled access to a global audience, enabling them to amplify their reach beyond traditional marketing channels.
2. **Cost-Effectiveness:** Social media marketing is often more cost-effective than traditional advertising methods, particularly for start-ups and small businesses with limited budgets. It allows technopreneurs to achieve substantial exposure with minimal financial investment.
3. **Targeted Advertising:** Social media platforms offer advanced targeting options, allowing technopreneurs to tailor their marketing efforts to specific demographics, interests, and behaviours. This precision targeting increases the efficiency and effectiveness of marketing campaigns, resulting in higher conversion rates.
4. **Brand Building:** Consistent and strategic use of social media helps technopreneurs establish and nurture their brand identity. Technopreneurs can cultivate a loyal following and differentiate themselves from competitors by engaging with their audience, sharing valuable content, and showcasing their unique value proposition.
5. **Customer Engagement:** Social media facilitates direct communication between technopreneurs and their customers, fostering deeper relationships and enhancing customer satisfaction. Technopreneurs can demonstrate their commitment to customer service and build trust with their audience by actively engaging with followers, responding to inquiries, and soliciting feedback.
6. **Data Analytics:** Social media platforms provide robust analytics tools that enable technopreneurs to track the performance of their marketing efforts in real time. By monitoring key metrics such as engagement, reach,



and conversion rates, technopreneurs can refine their strategies and optimise their campaigns for maximum impact.

7. **Adaptability:** The dynamic nature of social media allows technopreneurs to quickly adapt to changing market conditions and consumer preferences. By staying agile and responsive, technopreneurs can capitalise on emerging trends and capitalise on new opportunities.

3.5 Self-Assessment Exercise

1. How can social media marketing benefit my business?
2. What are the drawbacks of social media marketing?

3.6 Conclusion

In conclusion, social media marketing is an indispensable tool for technopreneurs seeking to establish a strong online presence, connect with their target audience, and drive business growth. By harnessing the power of social media effectively, technopreneurs can elevate their brand, expand their reach, and achieve sustainable success in today's competitive marketplace.

3.7 Unit Activities

Proposal Development: Based on their research, teams develop a comprehensive proposal that includes:

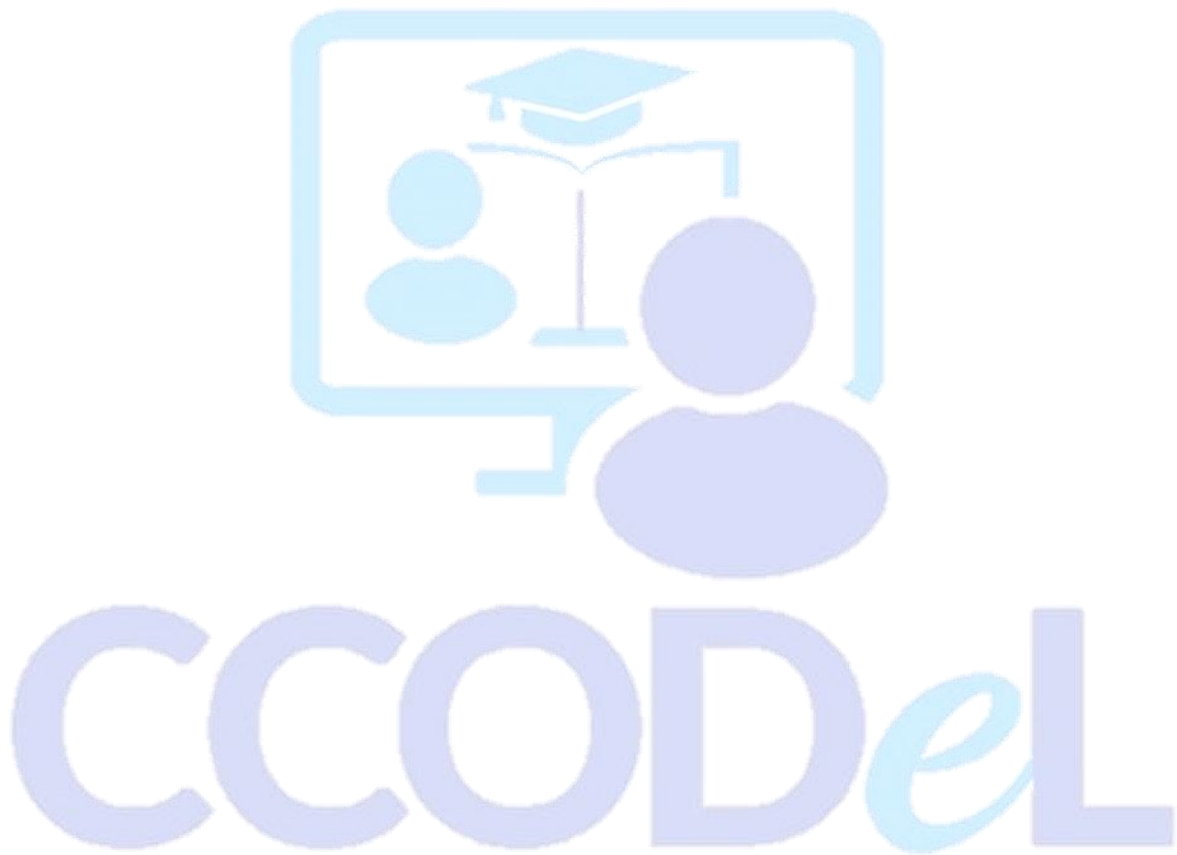
- Clear objectives and goals for their project.
- Strategies for integrating entrepreneurship, social media, and **cybersecurity**.
- Detailed plans for implementation, including resource allocation and timeline.
- Consideration of ethical and legal implications.
- Measures for evaluating the success of their proposal.

(Should be completed within a week)

3.8 Tutor-Marked Assignment



- What are some effective strategies for social media marketing in technopreneurship?
- Submit your responses to the dropbox for grading.





10.4 MODULE 5: CYBERSECURITY

10.4.1 UNIT 5: CYBERSECURITY

1.0 Introduction



In the modern world, there are always hazards and threats related to **cybersecurity**. Network and Internet infrastructure is becoming increasingly susceptible to a wide range of physical and cyberattacks. States and skilled cybercriminals use these vulnerabilities to steal money and information. Network security is the safeguarding of the network infrastructure from unauthorised access, misuse, or theft, as stated below:

- For several reasons, it is very challenging to safeguard our networks:
- Networks are becoming more intricate and interwoven.
- Physical devices are connected to networks.
- Networks are accessible to cybercriminals from anywhere in the globe.

1.0.1 Synopsis of the Domains in Cybersecurity

The various domains that comprise the "cyber world" are composed of numerous data groups. Groups start to gain power and influence when they can gather and use large volumes of data. Any kind of data that can be digitalised, including numbers, images, music, video, and photos, can be used as this data. These organisations might grow so strong that they function as independent entities, establishing distinct **cybersecurity** sectors.

Businesses like Google, Facebook, and LinkedIn could be seen as data domains in the digital era. To continue the analogy, one could argue that these digital enterprises' employees are cybersecurity professionals.

"Domain" is a word with numerous meanings. Anywhere there is protection, authority, or control.

2.0 Learning Outcomes



Cybercriminals novices or amateurs, sometimes known as "script kiddies," are unskilled individuals who frequently start assaults using pre-existing tools or online instructions.

At the end of this unit, you should be able to:

- define network security and its challenges
- recognise types of cyber criminals and tactics
- explain preventive measures against cyberattacks
- utilise frameworks for defining job roles
- appreciate the significance of professional certifications

3.0 Cybercriminals Against Experts In Cybersecurity

3.0.1 The Cybercriminals: Who Are They?

Early in the history of **cybersecurity**, most cybercriminals were either hobbyists or kids using a home computer to commit pranks and damage. The world of online thieves has grown increasingly perilous in recent years. Attackers are people or organisations that try to take advantage of weaknesses to benefit financially or personally. Credit cards, product designs, and anything else of value are all targets for cybercriminals.

Cybercriminals, novices, or amateurs, sometimes known as "script kiddies," are unskilled individuals who frequently start assaults using pre-existing tools or online instructions. While some want to show off their talents and hurt people, others are inquisitive. Even with simple tools, the outcomes can still be disastrous.

3.0.2 Cybercriminals

This gang of criminals breaches systems or networks to obtain access for various purposes. The intention behind the invasion determines whether to label these assailants as white, grey, or black hats. White hat Criminals break computer systems and networks in search of vulnerabilities that could be addressed to improve the security of these systems. In addition to receiving the assessment results, the device owners permit the infiltration.

On the contrary, black hat criminals take advantage of every possible flaw to further their illegal political, financial, or personal agendas. Grey-hatted attackers are positioned between white- and black-hatted attackers.



If a **vulnerability** is discovered, the grey hat attackers might notify the system owners if it aligns with their plan. Some grey hat hackers disclose its details online for other attackers to take advantage of the vulnerability.

3.0.3 Collective Cybercriminals

These criminals include terrorists, state-sponsored hackers, hacktivists, and groups of cybercriminals. Professional criminal groups that prioritise riches, power, and control are typically considered cybercriminals. The offenders are extremely well-organized and smart; they might even offer cybercrime as a service. Hacktivists make political comments to draw attention to topics that are significant to them. Hacktivists publicly publish embarrassing information about their victims. Attackers with state sponsorship obtain intelligence or carry out sabotage on behalf of their nation. Most of these attackers have extensive training and financial resources. Their attacks are concentrated on targets that are advantageous to their regime. Some terrorists with state sponsorship are even active military personnel in their home countries.

3.0.4 Motivations of Cybercriminals

Over time, the motivations and profiles of cybercriminals have evolved. Phreaking, also known as phone freaking, which is the practice of manipulating phone networks using different audio frequencies, is where hacking got its start in the 1960s. Criminals connected computers to networks using dial-up modems in the mid-1980s, and they employed password-cracking software to access data. Criminals these days aren't merely taking information. Malware and viruses are now considered high-tech weapons by criminals. Nonetheless, money is the primary driving force for the majority of cybercriminals.



Cybersecurity

organizations work together to create a database of vulnerabilities in order to restrict or counteract the flaws that cybercriminals may exploit.

3.1 Why Become an Expert in Cybersecurity?

More **cybersecurity** experts are in demand than there are in other IT positions. As technology changes the world and



elevates people's quality of life, it also leaves it more open to outside threats.

The following are the results of the lack of competent experts in **cybersecurity** and the level of competence needed for a successful **cybersecurity** expert:

- increased revenue possibilities and a demanding profession
- highly profitable career and service to the public
- Stopping Cybercriminals

Cybersecurity organisations work together to create a database of vulnerabilities to restrict or counteract the flaws that cybercriminals may exploit.

Among the measures implemented are:

- new legislation, early warning systems
- cyber intelligence sharing
- ISM requirements.

Creating guidelines for information security management in both domestic and foreign enterprises. One such example of these global initiatives is ISO 27000. Passing new legislation with harsh penalties to deter data breaches and cyberattacks.

3.2 Typical Dangers

Some experts are visionaries and inventors, as was previously mentioned. They construct many Internet cyber domains. They can identify and capitalise on the potential of data. After that, they establish businesses, offer services, and defend individuals against online threats. Professionals in **cybersecurity** should ideally be aware of the risk that data poses when it is misused against individuals.

Professionals in **cybersecurity** are primarily concerned with threats and vulnerabilities. Two circumstances in particular are crucial:

- When there is a chance that something bad, like an attack, will happen.



- When a target is exposed to an attack due to a **vulnerability**.

Criminals capture data streams across a network using packet sniffing tools. This implies that any private information, including credit card numbers, usernames, and passwords, is vulnerable. Packet sniffers function by monitoring and logging every bit of data that passes over a network. Rogue devices, such as unprotected Wi-Fi access points, can also be used by criminals. When a criminal places this close to a public area, like a coffee shop, unwary people might sign on, and the packet sniffer will copy their data.



Cyberattacks such as Stuxnet during the past ten years have demonstrated that it is possible for a cyberattack to effectively disrupt or destroy vital infrastructures.

A domain name, such as www.facebook.com, can be translated into its numerical IP address using the Domain Name Service (DNS). A DNS server will query another DNS server if it is unsure of the IP address. A DNS resolver's cache is tampered with by the criminal through DNS spoofing, also known as DNS cache poisoning. These "poison" attacks use a flaw in the DNS software that directs traffic for a particular domain to the criminal's computer rather than the rightful owner of the domain.

Packets are used to carry data over a network like the Internet. Packet forging, sometimes called packet injection, interferes with a communication network by generating packets that appear to be an integral component of a conversation. A criminal can disrupt or intercept packets by using packet forging. Through this procedure, a criminal might deny someone access to specific network services or hijack an authorised connection. This type of assault is known as a man-in-the-middle (MITM) attack.

Cyberattacks such as Stuxnet during the past ten years have demonstrated that it is possible for a cyberattack to effectively disrupt or destroy vital infrastructures. The SCADA, or supervisory control and data acquisition technology, tracks and handles the industrial processes. The Stuxnet attack was primarily directed towards this. SCADA is a component of many industrial communications, energy, manufacturing, and



production operations. To see more details regarding the Stuxnet attack, [Go to this URL](#).

3.3 Increasing Risks To Cybersecurity

Dangers to Internal Security: Both external and internal threats

Attacks can originate from the outside as well as the interior of an organisation, as the picture shows. Internal users, such as employees or business associates, may intentionally or unintentionally:



Both novices and experts can obtain entrance through external threats by using social engineering techniques like deception or vulnerabilities in networked devices.

- Confidential data mishandled
- Internal servers or network infrastructure equipment put in danger of failure
- By plugging infected USB media into the company computer system, you enable external attacks
- Intentionally allow malware to enter the network via phishing emails or internet pages

Since employees have immediate entry to the building's facilities, inside risks can cause more harm than exterior risks. Insiders are usually quite familiar with the company's network, assets, and private information. Regulations, higher administrative privilege levels, and security countermeasures may also be known to them.

Threats to External Security

Both novices and experts can obtain entrance through external threats by using social engineering techniques like deception or vulnerabilities in networked devices. External assaults leverage weaknesses or vulnerabilities to obtain internal resources.

Big Data's Effects

Large and complicated data sets are the source of big data, rendering conventional data processing tools insufficient. Three dimensions of big data present potential as well as challenges:

- The quantity or volume that contains data
- velocity or speed of data in question



- The spectrum or diversity of data sources and classifications

There are numerous instances of significant business hacking in the press. Media coverage of attacks on major corporations such as PayPal, Home Depot, and Target is substantial. Therefore, enterprise systems require substantial changes to security product designs and considerable enhancements to technology and operations. Governments and corporations are also passing new rules and regulations that demand stronger data security and protection procedures to better protect big data.

3.4 Developing Additional Experts to Solve the Cybersecurity Expert Shortage

The National Institute of Standards and Technologies, abbreviated NIST, created a structure for US businesses and associations requiring security experts. Companies can use the framework to determine the primary duties, job titles, and workforce skills required. **Cybersecurity** work is categorised and described using the National **Cybersecurity** Workforce Framework. It offers a standard vocabulary for defining **cybersecurity** jobs and a standardised list of duties and competencies needed to become a **cybersecurity** professional. The framework aids in defining **cybersecurity** professional requirements.

Cybersecurity job is divided into seven categories according to the Workforce Framework:

1. Operate and Maintain entails providing the administration, maintenance, and support needed to guarantee the security and performance of IT systems.
2. Threats to internal systems and networks are identified, analysed, and mitigated as part of the Protect and Defend process.
3. Investigate also includes looking into cybercrimes and/or cyber occurrences that involve IT resources.



4. Specialized deception and denial of service attacks, as well as the gathering of cybersecurity data, are included in the "Collect and Operate" strategy.
5. Incoming **cybersecurity** information is subjected to a highly skilled review and evaluation as part of the analysis process to ascertain its intelligence value.
6. Leadership, management, and guidance are provided via oversight and development in order to carry out **cybersecurity** work efficiently.
7. Developing and constructing secure IT systems are all part of securely provisioning.

Every category has a number of specialisation areas. Next, common **cybersecurity** work kinds are defined by the specialisation areas.

3.4.1 Certifications For The Industry

Information security specialists with the necessary skills and knowledge are desperately needed in the realm of **cybersecurity** threats.



Information security specialists with the necessary skills and knowledge are desperately needed in the realm of **cybersecurity** threats.

The IT industry established requirements for **cybersecurity** practitioners to obtain professional certifications attesting to their degree of expertise.

Obtain CompTIA Security+

CompTIA sponsors the Security+ exam program, which certifies IT administrators' proficiency in information assurance. The Security+ exam addresses cloud computing-related issues as well as the key concepts for network security and risk management.

Certified Ethical Hacker (CEH) by EC-Council

The **cybersecurity** specialists who obtain this intermediate-level certification are said to be knowledgeable and skilled in a variety of hacking techniques. To find weaknesses in systems and avenues of entry, these **cybersecurity** experts employ the same methods and abilities as online criminals.

Security Essentials for GIAC (GSEC) by SANS

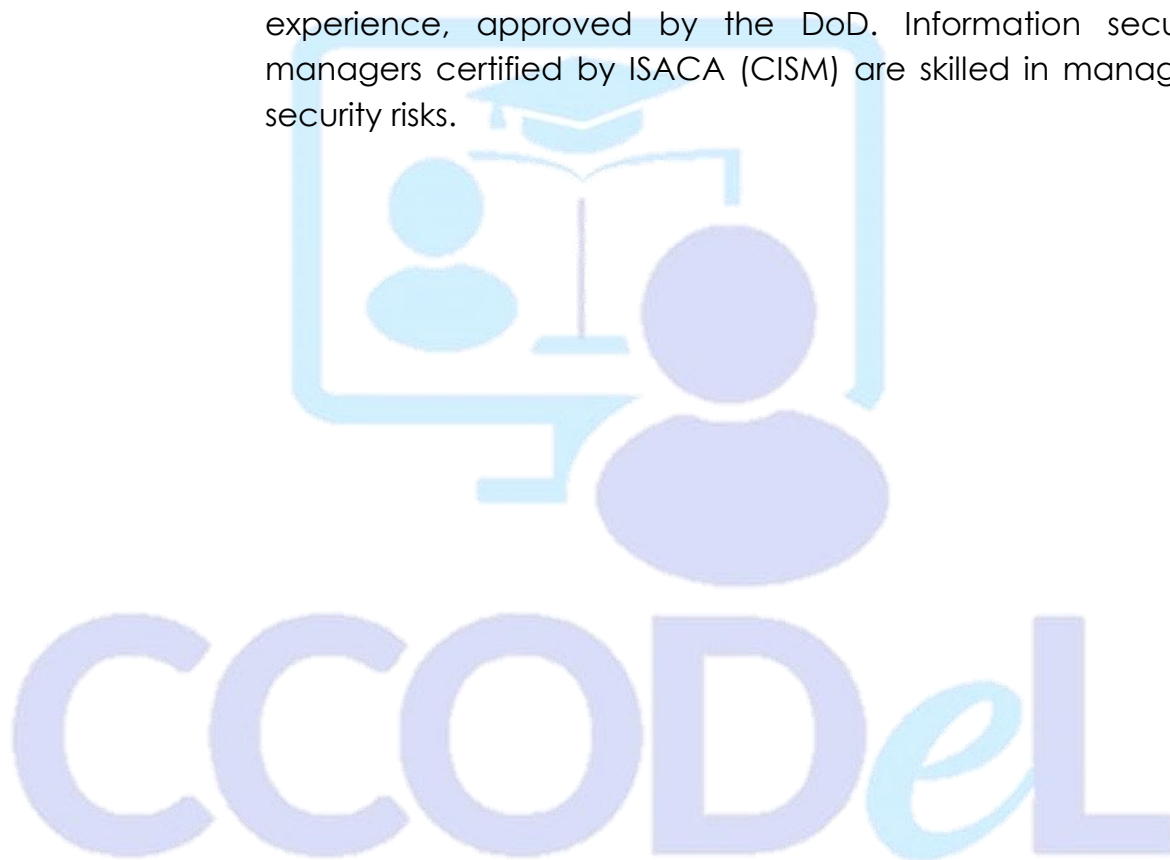
The GSEC certification is a good choice as an entry-level credential for those in the **cybersecurity** industry who can



demonstrate that they understand security terms and concepts as well as the skills and experience needed to perform direct security work. Numerous additional certificates in the domains of auditing, forensics, and security administration are available through the SANS GIAC program.

(ISC)² Certified Professional in Information Systems Security (CISSP)

The CISSP is a globally recognised credential for **cybersecurity** specialists with extensive technical and management experience, approved by the DoD. Information security managers certified by ISACA (CISM) are skilled in managing security risks.

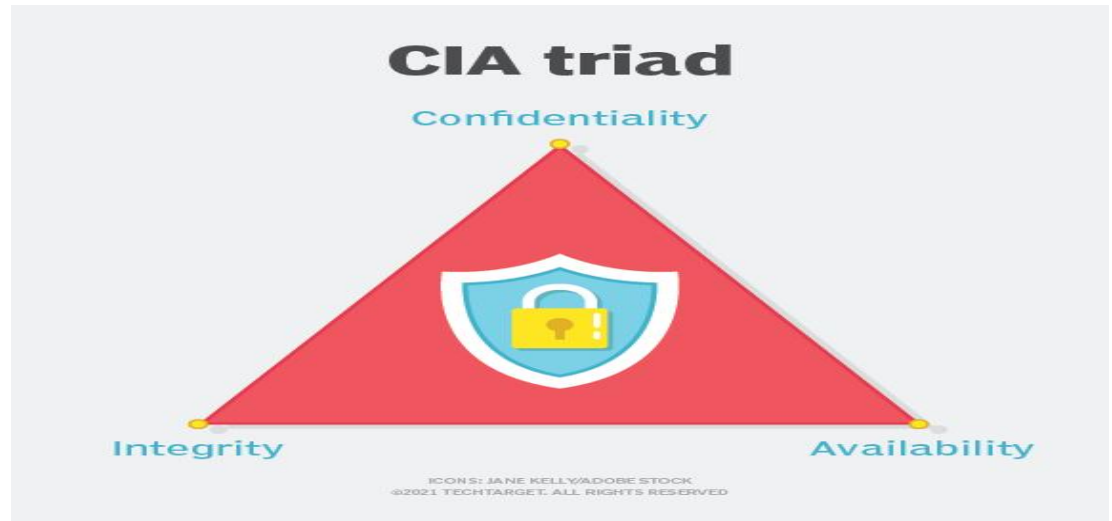




10.5 MODULE 6: THE CIA TRINITY: AVAILABILITY, INTEGRITY, AND CONFIDENTIALITY

10.5.1 UNIT 6: THE CIA TRINITY: AVAILABILITY, INTEGRITY, AND CONFIDENTIALITY

1.0 Introduction



1.0.1 What Is The Availability, Integrity, And Confidentiality?

Information security (InfoSec) policy inside a corporation might be guided by confidentiality, integrity, and availability, which is the first dimension of the **cybersecurity** cube.

1.0.2 The State of Data Information

Significant amounts of data, including payment processing systems, databases of consumers, records, messages, and medical information, are widely available online.

Cybersecurity experts focus on protecting these diverse types of data. The challenges of safeguarding all data in cyberspace are the subject of the state of data, which is the second dimension of the **cybersecurity** cube.

Three states are possible for data:

- Data stored or at rest
- Data in transit



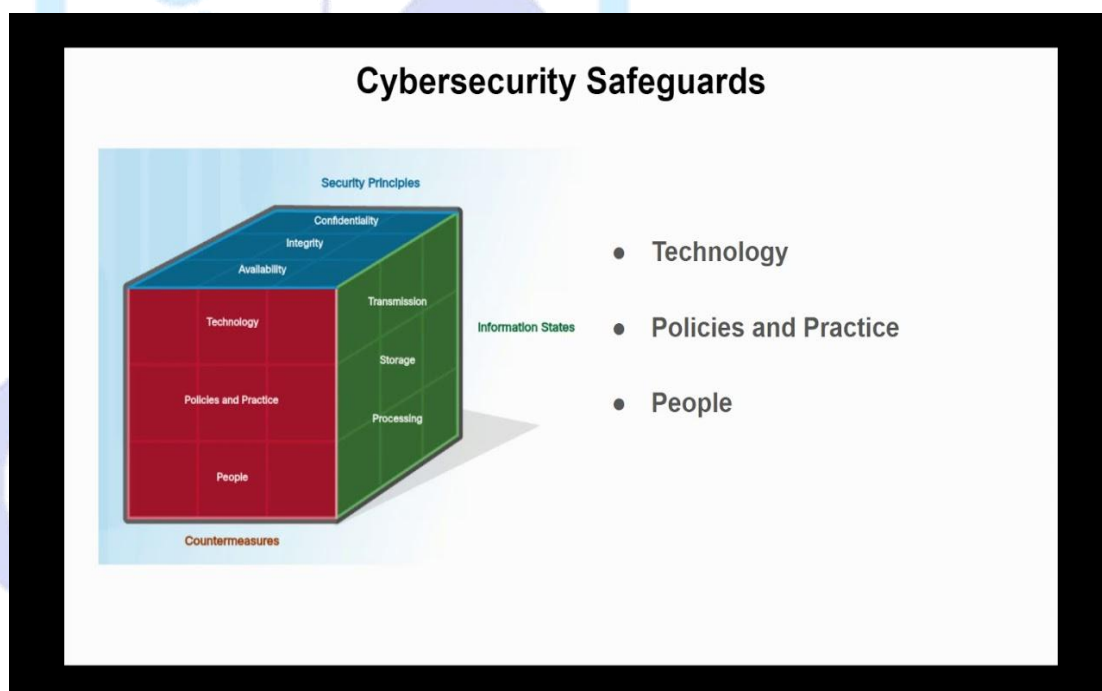
- Processed data

Cybersecurity experts must take into consideration the data security measures in each of the three states in order to defend cyberspace.

1.0.3 Protections Against Cybersecurity

The abilities and discipline that a **cybersecurity** specialist can use to safeguard cyberspace are outlined in the third level of the **Cybersecurity** Cube. When it comes to safeguarding data in cyberspace, **cybersecurity** experts need to employ a variety of diverse abilities and fields. To do this, they must continue to be on the "appropriate part" of the legal system.

The **Cybersecurity** Triangle illustrates the three basic skill sets and practices that are used to provide protection.



A three-dimensional **cybersecurity** cube image.

1.0.4 The Safeguards of Cybersecurity

These safeguards involve the three dimensions of the **Cybersecurity** triangle, namely:

- Security principles



- Information state
- Countermeasures

1.0.5 Limiting Access

Many security techniques described in access monitoring prevent unauthorised parties from accessing a computer, network, database, or other data resource. The AAA concepts encompass three security services: authorisation, authenticity, and auditing. These services provide the basic framework for access control.

The purpose of verification is to confirm a user's identity and stop unwanted access. Users use their ID or log in details to verify their personality. Furthermore, individuals must provide any of these details to confirm their identity:

- Anything they are aware of (password, for example)
- Anything they possess (like a card or token)
- What they are (like a fingerprint)

Authorisation solutions regulate the facilities that users can access and what tasks they can perform.

1.0.6 Techniques for transmitting information

Transmitting data between devices is referred to as data transmission. There are many methods for moving data between devices, including:

- Sneaker net: a technique for physically moving data by removable media between systems
- Networks that are wired, meaning they send data over cables
- Radio signals are used by wireless communication systems to transport data.

1.0.7 Crucial Cybersecurity Steps

You must implement the required security measures in order to shield your systems and data from online threats. These precautions include:

- Maintenance of infrastructure
- Updates for operating systems and devices



- testing of storage
- planning for crises
- implementation of new innovation
- observation of abnormal operations
- testing of resilience

2.0 Attacks, Vulnerabilities, and Threats to Cybersecurity

For **cybersecurity** experts, threats, vulnerabilities, and assaults are the **main** concerns. A threat is the potential for something bad to happen, like an attack. A flaw that leaves a target open to attack is called a **vulnerability**. An intentional exploitation of a **vulnerability** is called an attack.



For **cybersecurity** experts, threats, vulnerabilities, and assaults are the **main** concerns.

Any aggressive action performed by cyber criminals to target computer networks, computer information systems, or other computer equipment is referred to as a cyberattack.

2.0.1 What Is Malicious Software?

Malicious software, sometimes known as virus infections, is designed to disrupt computer functions or gain entry to networks against the user's awareness or consent.

Software infections include ransomware, spyware, adware, Trojans, horses, scareware, worms, and other harmful programs on computers. Viruses can range from being clearly obvious and simple to identify to being incredibly elusive and challenging to locate.

2.0.2 Viruses

A virus is an executable file that contains code that is maliciously linked to an approved software or another executable file. The majority of viruses can activate on a specified day or time and require end-user involvement. Three main channels are typically used by computer viruses to spread: portable media, Internet downloads, and email attachments. Viruses can be benign and only show an image, or they can become mutants and cause destruction by altering or erasing data. To evade being discovered,



Opening a file can be enough to start a virus. A file system virus, often known as a boot sector, can infect USB flash sticks and then move them to the computer's hard drive. An application virus can be activated by running a certain program. Once a program virus is operational, it typically spreads to other computers on the network or to other programs on the same machine. One infection that propagated over email was the Melissa virus. Tens of thousands of users were impacted by Melissa, who is estimated to have caused \$1.2 billion in damages. [Go to this URL](#) to learn more about Melissa.

Worms are dangerous programs that spread by themselves by taking advantage of holes in networks. Networks are often slowed down by worms. Worms can operate independently of a host program, but viruses need that application to function. Worms no longer require user involvement, aside from the initial infection. Once a worm has infected a host, it can quickly propagate throughout the network. Worms exhibit comparable behaviours. They all have a payload, a means of self-propagation, and an enabling **vulnerability**.

Worms are responsible for some of the most damaging cyberattacks. For example, in 2001, the so-called Code Red worm managed to infiltrate 658 servers. In only nineteen hours, the worm infected more than three hundred thousand servers.

2.0.3 The Virus Known As The Trojan Horse

Trojan horses are dangerous programs that carry out illegal activities while posing as desired ones, like playing games online. This malware makes use of the user's permissions. A Trojan horse differs from a virus in that it affixes itself to non-executable items, like computer games, music files, or graphic files.

2.0.4 The Computer Malware Known As Logic Bombs

A program that is malicious and utilises a trigger to start its harmful code is known as a logic bomb. Triggers might include things like dates, hours, running other applications, or the deletion of a user identity. Up until that point in time, the



logic bomb is dormant. A logic bomb can harm a computer by implementing harmful code once it is activated. A logic bomb can target operating systems or applications, destroy files, and tamper with database entries. Experts in **cybersecurity** have lately found logic bombs that target and damage a workstation or server's hard drives, CPU, memory, cooling fans, and power supply, among other hardware components. These gadgets are overdriven by the logic bomb until they malfunction or overheat.

2.0.5 Ransomware

Until the target pays, ransomware enslaves a computer system or the data within it. Typically, ransomware encrypts user data on the computer using a secret key. To get the restriction lifted, the user has to provide the thieves with a ransom.

Certain system vulnerabilities can be exploited by certain ransomware variants to lock down the system. Ransomware arises from a downloaded file or a software flaw and spreads like a Trojan horse.

The criminal always aims to make payments using an untraceable method. The thief provides a program that decrypts the files or delivers an unlock code once the victim makes a payment.

2.0.6 Rootkits and Backdoors

The program or code that a hacker introduces to gain access to a system is known as a backdoor. By passing the standard authentication procedure, the backdoor gains access to a system. Back Orifice and Netbus are two popular backdoor tools that let unauthorised individuals access systems remotely. Even if the business resolves the initial **vulnerability** used to attack the system, the backdoor's goal is to give cybercriminals access to the system in the future. Typically, to install a backdoor, authorised individuals unintentionally run a Trojan horse program on their computer.

A rootkit creates a backdoor by altering the operating system. After that, attackers can remotely access the machine by



using the backdoor. The majority of rootkits use software flaws to change system files and escalate privileges. By using programming mistakes or design defects, privilege escalation allows a criminal to gain higher access to network resources and data. In order to make themselves more difficult to find, rootkits frequently alter system forensics and monitoring software. A rootkit-infected machine frequently has to have its operating system completely removed and reinstalled.

2.1 Guarding Against Trojan Horses

All types of malware can be defended against with a few easy steps:

- **Antivirus Software:** Most antivirus programs detect and remove the majority of common malware types. On the other hand, cybercriminals create and launch new dangers every day. Thus, maintaining the most recent versions of the signatures is essential for a successful antivirus program. A fingerprint and a signature are similar. It recognises the traits of a malicious piece of code.
- **Update Software:** Numerous malware types accomplish their goals by taking advantage of holes in software, which can be found in both operating systems and apps. Even though operating system flaws were the primary cause of issues in the past, application-level flaws now represent the biggest danger. Regretfully, although operating system vendors are increasingly accommodating to patching, the majority of application suppliers do not share this sentiment.

2.1.1 Spam

Billions of people use email, making it a ubiquitous service. Email, one of the most widely used services, is now a significant security risk for both individuals and businesses. Spam is unsolicited email, sometimes referred to as junk mail. Usually, spam is employed as a marketing strategy. However, spam can also spread viruses, bad connections, or misleading information. The ultimate goal is to obtain personal information, such as bank account information or social



Spam is unsolicited email, sometimes referred to as junk mail.



security numbers. Most spam comes from multiple computers on networks that have been infected by viruses or worms. These compromised systems attempt to send out as much bulk email as they can.

Certain spam may still pass through despite the implementation of these security measures. Keep an eye out for some of the most typical signs of spam:

- There's no subject line on an email
- An email asking for an account update has been received.
- There are misspellings and odd punctuation in the email text.
- The email contains lengthy and/or mysterious links.
- An email appears to be correspondence from a reputable company.
- The user is asked to open an attachment in the email.

A user should refrain from opening an email or any attachments if they come across one or more of these symptoms. Communication policies within a business often require users to forward this type of information to the computer security department as soon as they get it. Nearly every email provider filters out spam. Regrettably, spam continues to use bandwidth and requires the recipient's server to process the message.

2.1.2 Phishing

Phishing is an art of deception. Internet criminals use messaging apps, email, and various online platforms to get information, including login details and account data, by impersonating reliable companies or people. The act of an adversary passing across a fake email as coming through a trustworthy and legitimate place is known as phishing. The text aims to trick the recipient into sending sensitive private or financial data or running malware on the device they are using.



A very focused phishing attempt is called spear phishing. Spear phishing targets a specific individual with personalised emails, whereas messages are used in spear phishing and phishing attacks to stay in connection with those they have hacked. Prior to sending the email, the criminal finds out what interests the target. For instance, a thief discovers that the victim has a passion for automobiles and has been eyeing a certain make and model. The offender writes the target an email, forges a car sale offer, and signs up for a similar car conversation group in which the victim participates. The email contains an entry to images of the car. The target unintentionally installs malware on the PC when they open the link.

2.2 Protecting Against Browser and Email Attacks

Using host/server filters, teaching users to be wary of unexpected emails, and filtering emails are some strategies for combating spam.



Employers also need to warn staff members about the risks involved in opening email attachments that can be infected with a virus or worm. Even when an email attachment appears to be from a reliable source, you should never assume that it is secure.

While eliminating spam is a challenging task, there are strategies to lessen its impact. For instance, before spam reaches a user's mailbox, the majority of ISPs screen it. Email filtering is automatically done by a lot of antivirus and email software products. In other words, they identify and eliminate spam from an email account.

Employers also need to warn staff members about the risks involved in opening email attachments that can be infected with a virus or worm. Even when an email attachment appears to be from a reliable source, you should never assume that it is secure. It's possible that a virus is attempting to spread via the sender's PC. Prior to opening an email attachment, always scan it.

An industry group called the Anti-Phishing Working Group (APWG) is dedicated to eradicating fraud and identity theft brought on by phishing and email spoofing. Maintaining the most recent security patches installed on the system to eliminate known vulnerabilities is ensured by keeping all software updated.



2.2.1 Using Social Engineering Strategies

Social engineering provides a completely non-technical means of allowing an intruder to find out more information about an individual. "Social engineering" is an assault technique used to force individuals to divulge personal information or do activities.

Social engineers take advantage of people's vulnerabilities and frequently rely on their want to assist others. An attacker might, for instance, phone a designated staff member with an urgent issue requiring instant access to the network. The attacker may play on the worker's conceit and use name-dropping to establish authority.

Some examples of social engineering attacks are as follows:

- Pretexting: The practice of an attacker calling a person and telling them a falsehood in an effort to obtain sensitive information. As an illustration, consider the case of an attacker who poses in need of financial or personal information to verify the recipient's identity.
- Deception

2.2.2 Strategies for Social Engineering

Social engineers use a variety of strategies. Among the social engineering techniques are:

- Authority: Individuals are more inclined to follow orders from "an authority"
- Intimidation: Criminals use intimidation to force a victim to act
- Consensus/social proof: People act if they believe that others will follow suit.
- Urgency: Individuals take action when they perceive a limited quantity of goods exists.
- Scarcity: individuals take action once they think there is a finite amount.

The first two types of relationships that criminals establish with victims are familiarity and liking. The second type is trust, which takes longer to develop.



It is the duty of **cybersecurity** specialists to inform other members of the organisation about social engineering techniques.

Email, Web-based, and Online Deception

Sending jokes, amusing films, and non-work-related emails to co-workers may be against the company's acceptable use policy and subject to disciplinary action.

Protecting Against deception

Companies must raise staff knowledge of social engineering techniques and provide them with the necessary training on preventative techniques, such as the following:

- Never give personal information or login credentials to strangers over the phone, in person, via chat, or by email.
- Avoid clicking on tempting links in emails and on websites. Be aware of unintentional or automatic downloads.
- Create guidelines and inform staff members of them.
- Provide workers with a sense of security ownership.
- Resist the urge to give in to pressure from strangers.

Denial of Service

A denial-of-service, usually abbreviated as DoS, interferes with the network operations that customers, gadgets, or applications may access.

Major DoS attack categories are as follows:

- **Overwhelming Volume of Traffic:** The attacker sends data at a rate that is too fast for the host, network, or application to handle. This results in a device or service crash, a lag in transmission or reaction, or both.
- **Maliciously Formatted Packets:** When an attacker sends a host or application a maliciously formatted packet, the recipient is unable to process it. An application, for instance, is unable to recognise packets that have defects or that the attacker has wrongly structured and routed. The receiving equipment operates very slowly as a result of this or falter.



Distributed denial of service attacks are harmful because they can rapidly and cheaply stop the flow of information, wasting an enormous amount of money and time. Even a novice attacker may carry out these attacks rather easily.

A denial-of-service attack aims to prevent authorised users from accessing the network, therefore rendering the network inaccessible. (Recall the three fundamental security principles: confidentiality, integrity, and availability above).

Similar to a DoS attack, a distributed denial of service (DDoS) attack comes from several synchronised sources. A DDoS assault might go like this, for instance:

An assailant constructs a **botnet**, or network of compromised hosts, made up of zombies. Those who are infected are zombies. The zombies are controlled by the attacker through handler systems. More hosts are continually scanned by the zombie computers, which multiplies the zombie population. When prepared, the hacker directs the handler systems to launch a DDoS assault using the zombie **botnet**.

(DDoS) intrusion employs a large number of zombies to overload a target.

Detecting

Eavesdropping on someone is comparable to sniffing. It happens when hackers sniff every packet that comes through their network user interface, irrespective of permission. Criminals use a combination of software and hardware, or the two, to sniff networks. Sniffing can view all network traffic or focus on a particular protocol, service, or even character string, similar to a password or login. In addition to observing all traffic, some network sniffers also alter some traffic or all of it.

Sniffing has advantages as well. Sniffers can also be used by network administrators to troubleshoot other network issues, analyse network traffic, and find bandwidth problems.

To stop sniffers from getting onto the internal network, physical protection is crucial.



Spoofing

Spoofing, a form of impersonation attack, exploits a system's trust relationship with another system. If the authentication completed by one system is accepted by the other, the user signed into one may not have to go and repeat the process again to access the second system. An intruder can exploit this setup by delivering a packet to one system that appears to have come from a reliable system. Without requiring authentication, the targeted system can carry out the desired task because a trusted connection has been established.

There are various kinds of attacks known as spoofing.

MAC address spoofing is the practice of a single system receiving data packets using someone's system's **MAC** address. IP spoofing sends information from a fictitious source address in order to disguise its identity by forging **ARP** messages. The Address Resolution mechanism (**ARP**) is a data transmission mechanism that resolves IP addresses to **MAC** addresses. Through the use of forged **ARP** messages, **ARP** spoofing allows a criminal's **MAC** address to be connected to an authorised network member's IP address over a local area network (LAN).

IP addresses and domain names are connected via the Domain Name System (DNS). Through DNS server spoofing, an IP address under the criminal's control is changed to serve as the destination for a particular domain name.

Attack by an intruder in between networks (Man-in-the-Middle)

When an intruder launches an attack known as a man-in-the-middle (MitM) by eavesdropping on computer conversations in order to steal data travelling over a network, since the hosts are oblivious to the changes made to the messages, the criminal may also opt to alter the messages and spread misleading information between them. MitM makes it possible for a criminal to take over a device without the user's awareness.



- One variant of man-in-the-middle is called Man-In-The-Mobile (MitMo). MitMo seizes command of a mobile gadget. Attackers receive user-sensitive data from the compromised mobile device. An example of a MitMo-capable **vulnerability** is ZeuS, which enables attackers to surreptitiously intercept user-sent 2-step verification SMS texts. For instance, when a user creates an Apple ID, they have to enter a phone number that can receive SMS messages in order to get a temporary verification code that they can text to confirm the user's identity. Malware eavesdrops on this kind of communication and sends the data out to the perpetrators.
- Repeat assaults take place when an intruder records a portion of a chat that involves two devices and then sends the message repeatedly afterwards. Replay-based exploits overcome authentication processes.

Keystroke Monitoring: A program that monitors or captures keystrokes in an operating system. Keystroke monitors can be generated by hackers through software installed on an electronic device or hardware that is directly attached to a device. The criminal sets up the keyboard logger program to ensure the log file is delivered. Keystrokes entered into a log file have the potential to reveal usernames, passwords, visited websites, and other personal information. Keyboard loggers may be genuine for-profit programs. Key logger software is frequently purchased by parents to monitor their children's online activities and websites. Unauthorised key loggers can be found and eliminated by a variety of anti-spyware programs. Despite being legal, criminals use keylogging software illegally.

2.3 PROTECTING AGAINST ATTACKS

There are several actions an organisation can take to protect itself from different types of attacks. Firewalls should be set up to reject any packets coming from the outside that have addresses that point to the network as their source.

Preventing DDoS and DoS attacks:



- Make sure patches and upgrades are up to date.
- Divide the workload among server systems
- lock external Internet Control Message Protocol (ICMP) packets at the border. This situation is unusual, and it suggests that a cybercriminal attempted a spoofing attack. ICMP packets are used by network devices to transmit error messages. For instance, the ping command makes use of ICMP packets to confirm that a device on the network is able to connect with another.

By encrypting communication, offering cryptographic authentication, and appending a time stamp to every message segment, systems can block replay attacks.

WPA and WEP: A security protocol called cable Equivalent Privacy (WEP) aims to give wireless local area networks (WLANs) the same degree of protection as cable LANs. While physical security measures aid in the protection of a cable LAN, Wireless Ethernet Protocol (WEP) aims to offer comparable security for data transferred over a WLAN through encryption. A key is used by WEP to encrypt data. Since WEP does not offer key management, the number of users sharing the key will only increase. Because all users are utilising the same key, an attacker can access a lot of traffic to use in analytic attacks.

Additionally, WEP's initialisation vector (IV), one of the cryptographic system's components, has a number of issues:

- The field size is only 24 bits, which is insufficient.
- It is readable because it is clear text.
- Since it is static, on a busy network, identical key streams will recur.

The enhanced Wi-Fi Protected Access (WPA) and later WPA2 protocols were released to replace WEP. WPA2's encryption issues are not the same since an intruder cannot figure out the key by watching traffic. Because hackers can examine the packets passing between an authorised user and the access point, WPA2 is vulnerable to attack. After using a network sniffer, cybercriminals assault the passphrase offline.



By encrypting communication offering cryptographic authentication, and appending a time stamp to every message segment, systems can block replay attacks.



A passphrase is basically a longer password, usually at least 14 characters in length, with spaces between words. Both passwords and passphrases can be used to encrypt data and maintain secure access to websites, software, and hardware systems.

Protecting Against Attacks Using Wireless and Mobile Devices

To protect yourself against wireless and mobile device attacks, adopt a few precautions. Most WLAN equipment uses default settings. To utilise the essential wireless security features, such as authentication and password protection, modify the default configuration settings.

Access points can be placed inside a security perimeter (a restricted zone), which is also where other untrusted devices like web servers and email reside or within a firewall. Unauthorised workstations or rogue access points may be found using WLAN programs like Nets tumbler. To deal with circumstances when approved guests need access to the Internet while they are visiting, develop rules and regulations for your visitors to use the Internet. If employees are permitted to use WIFI networks, use a VPN, or virtual private network, to provide a remote connection.

Cross-Site Scripting

XSS, or cross-site scripting, is an attack that can affect online applications. Criminals can add codes to online resources that users visit using the XSS technique. This application might contain dangerous code. Cross-site scripting involves the following groups: the web page, the adversary, and the person being targeted. The perpetrators do not particularly target the victim. The adversary exploits the weakness of a web page or web service. Individuals who visit websites that have adversaries or hackers install client-side scripts on them are considered victims. Unknowingly, the dangerous script makes its way to the user's browser. This kind of malicious software can access any cookies, session tokens, or private data. The victim's session cookie can be used by thieves to assume the identity of that user.



Injection of Code

Using a database is one method of storing data on a website. Extensible Mark-up Language (XML) and Structured Query Language (SQL) databases are two examples of the various types of databases. SQL injection and XML injection attacks make use of application flaws such as improper database query validation.

Injection of XML

An intrusion known as an XML injection occurs when data is used in an XML format database. The system queries to obtain the necessary data when the user enters it. The issue arises when the system does not thoroughly examine the user's input request. Criminals have the ability to access the database's contents and modify the query to fit their demands.

The thieves have access to all sensitive data kept in the database and can alter the website in any way they see fit. An XML injection attack puts the website's security at risk.

SQL Injection

The cybercriminal inserts a malicious SQL statement into an entry field to take advantage of a **vulnerability**. Once more, the system does not appropriately filter the user input for characters found in an SQL statement. SQL injection is a tool used by fraudsters on websites and SQL databases.

Criminals have the ability to assume the identity of someone else, alter or remove data, or take control of the database server.

In-Text Question (ITQ)

What role does technopreneurship play in addressing the **cybersecurity** skills shortage?

In-Text Answer (ITA)

Technopreneurship plays a crucial role in addressing the **cybersecurity** skills shortage by promoting education, training, and mentorship programs to develop the next generation of **cybersecurity** professionals. Technopreneurs create



opportunities for aspiring **cybersecurity** professionals to gain hands-on experience, acquire relevant skills, and contribute to the development of innovative solutions that strengthen cyber defence capabilities

2.4 How to Keep Information Safe



The study of creating and deciphering secret codes is known as cryptology. Creating and utilizing codes is known as cryptography.

The way that contemporary protocols and algorithms safeguard communications is explained by the concepts of cryptology. The study of creating and deciphering secret codes is known as cryptology. Creating and utilising codes is known as cryptography. Cryptanalysis is the study of and cracking of codes. For millennia, people have utilised encryption to safeguard confidential documents. For instance, Julius Caesar encrypted communications to his generals in the field using a straightforward alphabetic cypher. The cypher key needed to decode the messages would be known to his generals. Secure communications are now guaranteed by contemporary cryptography techniques.

What Is Meant By Cryptography?

The study of creating and deciphering secret codes is known as cryptology. Data can be transmitted and stored using cryptography so that only the intended receiver can access or process it. Computationally secure techniques are used in modern encryption to ensure that protected data cannot be readily compromised by cybercriminals.

Data secrecy protects privacy by limiting message reading to the intended recipient. Parties use encryption to do this. Data is jumbled during the encryption process to make it difficult for an unauthorised party to read.

Encrypted text, also known as cypher text, is encrypted text; readable data, on the other hand, is plain text, or clear text when encryption is enabled. A readable message in plaintext is changed into an unreadable, disguised message called cypher text through the process of encryption. The process is reversed by decryption. A key is also necessary for encryption, and it's essential to both encrypt and decode messages. The



cypher text can be decrypted to plaintext by the owner of the key.

Parties have employed a variety of encryption techniques and algorithms in the past. The method or technique used to solve a problem is called an algorithm. It is alleged that Julius Caesar encrypted messages by aligning two letter sets side by side and then moving one of them a predetermined distance. The key is the amount of places in the shift. With this key, he transformed plaintext into cypher text; only his generals, who also possessed the key, could decode the dispatches. The Caesar cipher is this technique.

Cryptography's Past

Thousands of years ago, in diplomatic circles, cryptography first came into existence. A monarch's court sent emissaries to other courts with encrypted communications. From time to time, courts that were not parties to the correspondence tried to intercept dispatches intended for a kingdom that they viewed as an enemy. Military leaders soon began utilising encryption to protect messages.

There are two types of encoding: digital encoding, which combines several techniques and procedures to guarantee confidentiality.

Encoding provides two means of ensuring data safety:

- Safeguarding the encryption process is the first encoding method. Protecting the encoding process procedure is essential if its secrecy is what keeps the system secure. If the method's details are discovered every time, all parties affected would be forced to modify it, which would not sound very safe or sustainable.
- Keeping your credentials guarded is the next approach. The techniques applied in the latest encryption technology are made available to the public. The data's confidentiality is guaranteed by the cryptographic keys. Passwords and the data that has to be encrypted are inputs into an encryption method coupled with cryptographic keys.

**Two categories of encryption algorithms exist:**

The pre-shared key, also referred to as a secret key pair, is used by symmetrical processes to encode and decode data. The sender and the person receiving the message know the pre-shared key before any encrypted communication occurs. The original text is encrypted and decrypted using the same key using symmetric methods. Usually, utilised key encryption methods need less processing power and are simpler to grasp.

Asymmetric procedures: these encryption procedures use one key for encoding data and a separate key for data decoding. One of the keys is open to everyone, and the other is confidential. Messages sent with a public-key security system are protected with the receiver's personal key; only the receiver of the message can decode the message using his personal key. Parties exchange encrypted communications without the need for a pre-shared key. Asymmetrical procedures are more complex, take longer to execute, and consume lots of resources.

Methods of Symmetric Encryption

Symmetric algorithms also referred to as private-key encryption, encrypt and decrypt data using the same pre-shared key.

As an illustration, despite living in distant places, Segun and Tola want to use the mail system to send each other vital messages. Segun wants to discreetly communicate with Tola.

Symmetric algorithms are used in private-key encryption. Segun and Tola share the same keys for a single padlock; before any confidential communications were sent, there was a key exchange. Segun uses the padlock to secure a little box in which he has written a hidden message. He sends Tola the box via mail. As the box travels via the post office system, the message within is secure. Tola uses his key to open the padlock and get the message when he gets the box. Tola can secretly reply to Segun using the same box and padlock.

Tola needs a new pre-shared key to communicate with Tope and keep it hidden from Segun. Segun will need to maintain



more keys the more individuals he wants to safely connect with.

Uses

As long as suitable methods are effectively used, encryption allows us to convert simple text into a cypher or content that is unreadable to hackers. In essence, approaches to encryption are intricate computations. Computing devices don't actually work with the alphabet and words. Rather, they operate using a binary structure, which consists of 1s and 0s. A collection of 8 bits, each of which is a 1 or 0, is referred to as a byte.

Here are a few instances of how it was used:

Mozilla Firefox, Microsoft applications, and MasterCard techniques for payments

Asymmetric key algorithms are used in four protocols.:

- One essential element of IPsec Virtual Private Networks (VPNs) is Internet Key Exchange (IKE).
- Secure Socket Layer (SSL), an add-on for web browsers that makes cryptography possible.
- One protocol that provides a secure connection for individuals to gain access to communication equipment is Secure Shell, or (SSH).
- An open network, usually online platforms, a virtual private network, or VPN, creates a secure communication channel. An Internet connection can be established between two endpoints, such as two distant offices, using a VPN connection. IPsec is used by VPNs. A group of protocols known as IPsec were created to provide secure network services. Access control, confidentiality, integrity, and authentication are all possible with IPsec services. Remote sites can communicate certified and encrypted data thanks to IPsec.

A rising number of businesses are concerned about data in use. Since the user must open and modify the data while it is



in use, it is no longer protected. Sensitive information, such as the encryption key, may be stored in system memory, which is used to store data in use. Criminals will have access to both data in motion and data at rest if they manage to breach data in use.

2.5 Self Assessment Exercise

1. How can you protect information from cybercriminals?

2.6 Conclusion

The CIA Trinity provides a comprehensive framework for addressing the fundamental objectives of **cybersecurity**: protecting information from unauthorised access, ensuring its accuracy and reliability, and maintaining its availability for legitimate users. By integrating measures to uphold confidentiality, integrity, and availability, organisations can effectively mitigate **cybersecurity** risks and safeguard their critical assets against a wide range of threats. However, it's essential to recognise that achieving these objectives requires a multi-layered approach, continuous monitoring, and adaptation to evolving **cybersecurity** threats and challenges.

2.7 Summary

The CIA Trinity, which stands for Confidentiality, Integrity, and Availability, forms the cornerstone of **cybersecurity** principles. Here's a summary and conclusion regarding each aspect:

1. **Confidentiality**: Confidentiality ensures that information is accessible only to those who are authorised to access it. This is typically achieved through encryption, access controls, and proper data handling procedures. By maintaining confidentiality, organisations protect sensitive information from unauthorised access, thereby preventing data breaches and safeguarding privacy.
2. **Integrity**: Integrity refers to the accuracy and reliability of data throughout its lifecycle. It ensures that information remains unaltered and trustworthy, both in storage and transmission. Techniques such as checksums, digital signatures, and access controls help maintain data



integrity by detecting and preventing unauthorised modifications. By upholding integrity, organisations can mitigate the risks associated with data tampering and **maintain** the consistency and reliability of their information assets.

3. **Availability:** Availability ensures that information and resources are accessible to authorised users whenever needed. This involves implementing measures to prevent and mitigate disruptions, such as system failures, cyberattacks, and natural disasters. Redundancy, backups, disaster recovery plans, and robust network infrastructure are key components of ensuring availability. By ensuring availability, organisations minimise downtime, maintain productivity, and deliver reliable services to users.

2.8 Unit Activities

Presentation and Peer Review: Each team should present their proposal to the class within the time frame.

2.9 Tutor-Marked Assignment

Complete assignment: Presentations and peer review.

CCODEL



10.6 MODULE 7: CONTROLS FOR PHYSICAL ACCESS

10.6.1 UNIT 7: CONTROLS FOR PHYSICAL ACCESS

1.0 Introduction



Real barriers are used as physical access restrictions to keep people from coming into direct touch with systems. Preventing unauthorised individuals from physically accessing buildings, machinery, and other organisational assets is the aim.

Physical access control establishes who may go in, how individuals may pass through, and at what times they can enter.

Physical security measures include items like:

- Barriers surrounding information technology facilities
- security devices, security officers,
- authorisation cards
- video surveillance equipment
- security alarm detectors

1.0.1 Access Controls With Logic

Logical access controls are the computers, equipment and application software that are used to govern who has access to what facilities and solutions. Computer systems use identification, authentication, authorisation, and accountability tools and standards as part of these technology-based solutions.

Some examples of logical access controls are as follows:

- Passwords are protected strings of characters.
- Encryption is the technique of converting plaintext into cypher text.
- Smart cards contain an implanted microchip.
- Biometrics are the physical traits of users.



- Routers connect at least two networks with protocols that are set up with rules that regulate data flow between devices;
- Firewalls block undesirable network traffic.
- Networks are monitored by intrusion detection systems for any suspicious activity.

1.0.2 Controls Over Administrative Access

Administrative access controls are procedures and policies created by organisations that are used to set up and monitor all parts of limiting unauthorised access. Administrative controls primarily address two areas: personnel and business operations.

The following are some of the administrative controls:

1. The organisation must develop policies and procedures to locate competent workers when hiring and do a background check. Criminal records checks are a type of employment screening that comprises inquiries into criminal histories, financial records, and past work history verification.
2. Security training and authentication: This informs staff members of the organisation's security standards. The most often used technique for authentication is the use of passwords. A password is a combination of characters that is used to authenticate a person. The word "password" can also refer to passphrase, passcode, passkey, or PIN. Cybercriminals will have an easier time guessing a user's password if this string of characters contains information that can be linked to the user, such as their address, birthdate, or name.
3. What you have: A password should include at least eight characters, according to a variety of publications. Users shouldn't make their passwords so short that they can be cracked by password thieves or so long that they are hard to remember. Numbers, special characters, and capital and lowercase letters should all be included in passwords.



4. Who You Are: Biometrics is the term for a distinct bodily trait used to identify an individual. In order to authenticate users, biometric security compares bodily attributes with profiles that have been stored. A profile is a data file that lists an individual's known attributes. If the user's attributes match the recorded settings, the system allows access. Among the common biometric devices is the fingerprint reader.

Two categories of biometric identifiers exist:

- Physiological traits, such as face, hands, fingerprints, DNA, retina, or ear features.
- Behavioural traits - these comprise habitual behaviours like voice, gestures, typing rhythm, or gait.

Presentation and Peer Review: Each team presents their proposal to the class. After each presentation, other teams provide constructive feedback and ask questions to deepen their understanding of the topic.

2.0 Learning Outcomes

At the end of this unit, you should be able to:

- Understand and implement core security principles, including multi-factor authentication, authorisation management, and implementation of security controls to mitigate risks effectively.
- Maintain data integrity through techniques such as salting, hashing, and digital signatures, ensuring reliable information management.
- Conduct thorough security assessments, analyse vulnerabilities, and develop proactive strategies for prevention, fostering a culture of continuous improvement in cybersecurity practices.

3.0 Authentication With Many Factors

At least two verification techniques are used in multi-factor authentication. An excellent example is a security key fob.





With multiple layers of authentication, the likelihood of identity theft can be decreased since hackers are unable to access personal information with just their username and password.

The two components are an owned item, like a security key fob, and something you know, like a password. Go one step further and incorporate an authentic element, like an image using a person's fingerprint.

With multiple layers of authentication, the likelihood of identity theft can be decreased since hackers are unable to access personal information with just their username and password. For instance, a password and PIN that the user receives on their smartphone may be needed to access an online banking page. Taking out cash from an ATM is another instance of multifactor authentication, as illustrated in the figure. Before the ATM can distribute money, the user needs to have a bank card and be aware of their PIN. Making Use of Authorization The first step in managing access is to define authorisation rules. These guidelines are established by authorised policies.

A policy on group membership establishes authorisation according to affiliation with a certain organisation. For instance, each employee at a company has a swipe card that allows them to enter the building. An employee's security card will prevent her from entering the server room if her work does not need her to have access to it.

Access permissions are defined by an authority-level policy according to an employee's position inside the company. For instance, the server room in an IT department is only accessible to senior-level staff members.

3.0.1 Accountability: What Is It?

Accountability collects usage data, reports it, and links certain behaviours to the individual or procedure that altered the machine setup. This information might be used by the company for invoicing or auditing. The information gathered may comprise the user's login time, the success or failure of the user's log-in, and the network resources the user visited. This makes it possible for a company to track actions, blunders, and errors made during an audit or probe.

3.0.2 Putting Accountability Into Practice



Accountability is implemented by rules, processes, technologies, and education. Log files offer comprehensive data according to the selected settings. An organisation might check the log to see which login attempts succeeded and which ones failed. Failures to log in may be a sign that someone attempted to hack an account. A company can learn which users are using what resources and when by looking at login success rates. Is it typical for a person with permission to log onto the company network at three in the morning? The policies and procedures of the organisation specify what activities need to be documented as well as how log files are created, examined, and kept.

Accountability is ensured by data retention, media disposal, and compliance requirements. A lot of laws mandate taking steps to secure certain kinds of data. An organisation is guided by these regulations for the proper handling, storing, and disposal of data. Accountability can also be enhanced by education and knowledge about an organisation's policies, practices, and relevant legal frameworks.

3.1 Control Mechanisms

3.1.1 What Are Security Controls Trying to Achieve?

As was previously noted, minimising weaknesses within an organisation is the primary goal of putting safety safeguards in place.

A security control's ability to be implemented effectively depends on how it is classified in relation to the security incident.

The following is a list of common classification kinds with a description for each:

- The goal of preventive controls is to stop an incident from happening.
- Detective controls make an effort to find occurrences after they've happened.



A security control's ability to be implemented effectively depends on how it is classified in relation to the security incident.



- The goal of corrective controls is to lessen the effects of an occurrence.
- The goal of deterrent controls is to dissuade people from starting an incident.

When a primary control is impractical, compensating controls are employed as an alternative.

TYPES OF SECURITY CONTROLS	CONTROL FUNCTIONS		
	PREVENTATIVE	DETECTIVE	CORRECTIVE
	PHYSICAL CONTROLS • Fences • Gates • Locks	• CCTV • Surveillance Cameras	• Repair physical damage • Re-issue access cards
	TECHNICAL CONTROLS • Firewall • IPS • MFA • Antivirus	• IDS • Honeypots	• Vulnerability patching • Reboot a system • Quarantine a virus
	ADMINISTRATIVE CONTROLS • Hiring & termination policies • Separation of duties • Data classification	• Review access rights • Audit logs and unauthorized changes	• Implement a business continuity plan • Have an incident response plan

3.1.2 Steganography

The technique of hiding data within an alternate communication or material item to prevent discovery is known as steganography. Steganography has an advantage over cryptography in that no special attention is drawn to the hidden message. Seeing the file in hardcopy or electronically would never reveal to anyone that an image truly held a secret message.

3.1.3 Techniques For Steganography

Least Significant Bits (LSB) is the method used to incorporate data in a cover image. Each pixel in the image is used in bits using this method. The fundamental building block of programmable colour in a computer image is a pixel. Three



colours are combined to create the unique hue of a pixel: red, green, and blue (RGB). A pixel's colour is specified by three bytes of data (one byte for each colour). A byte is made up of eight bits. All three bytes are used in a 24-bit colour scheme. LSB uses a small amount of the colour components red, green, and blue. Bits can be stored in each pixel.

Three pixels from a 24-bit colour picture are displayed in the figure. The letter T is one of the letters in the secret message, and adding the character T only modifies two shades of the colour. Even the smallest modifications to a bit are invisible to the human eye. A hidden character is what's left over. To effectively conceal a secret message, an image's bits usually need to vary by no more than half.

3.1.4 Identification

Steganalysis is the process of identifying hidden information. Steganalysis aims to reveal the information that is buried. The stego-image's patterns arouse suspicion. For instance, information may be hidden by unoccupied space on a drive. Tools for disk analysis can provide information about data that is hidden in underutilised storage device clusters. Filters can capture data packets with concealed information in the packet headers. Steganography signatures are used in both of these techniques.

3.2 The Craft of Maintaining Honesty

Integrity guarantees that during its whole life cycle, data is unaltered and reliable for everyone and everything. Data integrity must be given high importance in the design, implementation, and use of any device that handles, saves, or transmits data. The first section of this chapter covers the many forms of data integrity measures that are employed, including salting and hashing techniques. Users are able to verify the validity of data or messages. Digital signatures and certificates, which include data integrity measures, are used. A discussion of database integrity enforcement brings the chapter to a close. A database system's stability,



Data integrity must be given high importance in the design, implementation, and use of any device that handles, saves, or transmits data.



performance, and maintainability are all improved by having a well-defined and managed data integrity mechanism.

3.3 Conclusion

1. **Assessment Results:** Summarize the findings of the physical assessment. Highlight any vulnerabilities, weaknesses, or areas of concern that were identified.
2. **Impact Analysis:** Discuss the potential impact of the identified vulnerabilities on the overall security posture of the organisation. Consider how these vulnerabilities could be exploited by attackers and the potential consequences.
3. **Recommendations:** Provide recommendations for addressing the identified vulnerabilities and improving physical security measures. This could include suggestions for implementing new controls, enhancing existing security measures, or conducting further assessments.
4. **Risk Assessment:** Evaluate the level of risk associated with the identified vulnerabilities. Consider the likelihood of exploitation and the potential impact on the organisation's operations, assets, and reputation.

3.4 Summary

1. **Scope and Objectives:** Recap the scope and objectives of the physical assessment. Remind the reader of what was assessed and why it was important.
2. **Methodology:** Briefly outline the methodology used during the assessment. This could include details about the techniques, tools, and procedures employed.
3. **Key Findings:** Summarize the key findings of the assessment. Highlight the most significant vulnerabilities or weaknesses that were identified.
4. **Strengths:** Discuss any strengths or areas of good practice that were observed during the assessment. This could include aspects of physical security that are



working well and should be **maintained** or replicated elsewhere.

5. **Next Steps:** Outline the next steps that should be taken based on the findings of the assessment. This could include implementing recommended security controls, conducting further assessments, or reviewing and updating security policies and procedures.

6. **Conclusion:** Conclude the summary by emphasising the importance of physical security in the overall **cybersecurity** posture of the organisation. Reinforce the need for ongoing vigilance and continuous improvement in physical security measures.

By following this structure, you can effectively communicate the findings, recommendations, and implications of the physical assessment in the context of **cybersecurity**.

3.5 Self-Assessment

1. How can we prevent hackers from accessing our data or information?

3.6 Tutor Marked Assignment

1. **Presentation Preparation:** Teams prepare a presentation to communicate their proposal effectively. The presentation should be engaging, informative, and visually appealing.

2. Submit your responses for grading in the drop box. (Should be completed within a week).



10.7 MODULE 8: THE CODE OF ETHICS FOR CYBERSECURITY EXPERTS

10.7.1 UNIT 8: THE CODE OF ETHICS FOR CYBERSECURITY EXPERTS

1.0 Introduction



Conduct is the quiet language in the context that advises a person with expertise in **cybersecurity** on which actions to perform, regardless of whether it is legal. The **organisation gives cybersecurity experts access to the most private information and resources**. The **cybersecurity** expert must comprehend how moral judgments are influenced by the organisation's goals and the legislation.

Cybercriminals are acting unethically when they breach a system, take credit card details, and spread a worm. If the behaviours of a **cybersecurity** specialist are comparable, how is a company perceived? For instance, by patching a worm, a **cybersecurity** expert may be able to prevent it from spreading. The **cybersecurity** expert is essentially unleashing a worm. Does this scenario get a pass, though, as this worm is not malicious?

1.0.1 The Approach of Rights

The rights approach is based on the idea that everyone has the freedom to make their own decisions. This viewpoint considers how one's actions impact the rights of others while determining the morality of a certain course of action. These rights include the freedom to know the truth, to be alone, to be safe, and to have everyone treated equally by the law.

The Common-Good Method According to the Common-Good Approach, anything that is advantageous to the community is the common good. In this instance, a **cybersecurity** expert considers how a decision may impact the community's or society's overall welfare.

2.0 Learning Outcomes



At the end of this unit, you should be able to:

- Understand the concept and significance of hashing in **cybersecurity**, including its role in data integrity, authentication, and password protection.
- Recognise the importance of high-availability systems and their components in minimising downtime and ensuring continuous access to data and services.
- Develop practical skills in setting up a **cybersecurity** lab for ethical hacking purposes, including installing virtualisation software, configuring virtual machines, and using penetration testing tools.

3.0 Hashing: What Is It?

The hash tool ensures and verifies data integrity through the use of a cryptographic hashing function. Moreover, Data integrity must be given high importance in the design, implementation, and use of any device that handles, saves, or transmits data. Capable of authenticating. Since hash functions are one-way functions, they can be used in place of encryption keys or plain text passwords. This implies that the hash digest produced by a password hashed using a certain hashing method will always be the same. It is considered one-way, as it is technically impossible for two different bits of data to use hashing functions to generate the same compute digest. These values prevent unintentional or deliberate data alteration as well as unintentional data corruption. Furthermore, it is highly efficient hashing. A huge file or all of the data on a disk drive produces a hash value that is the same size.



The mathematical function of hashing is one-way, straightforward to compute, and very difficult to reverse.

3.0.1 Properties of Hashing

The mathematical function of hashing is one-way, straightforward to compute, and very difficult to reverse. One useful comparison for a one-way function is grinding coffee. Coffee beans are simple to crush, but reconstructing the original beans from all of the microscopic pieces is nearly hard.



The characteristics of a cryptographic hash function are as follows:

- The output has a set length, whereas the input can be any dimension.
- The process of the hashing procedure is irreversible and in a single direction.
- It's also uncommon for two separate inputs to yield identical encrypted values.

3.0.2 Hashing Functions

Hashing functions are useful in preventing unintentional data changes due to human error or communication problems. For example, a sender could wish to ensure that a message doesn't get altered while it's being delivered to the receiver. The message is entered into a hashing algorithm by the sending device, which then calculates the message's fixed-length digest or fingerprint.

3.0.3 Current Hashing Techniques

Nowadays, a wide variety of contemporary hashing algorithms are in use. The two most well-liked ones are SHA and MD5.

- **MD5, or Message Digest 5, algorithm**

128 bits is the hash value that the MD5 algorithm produces. In 2012, the security of MD5 was breached by the Flame malware. The Flame malware's creators forged a Windows code-signing certificate using an MD5 collision.

Any quantity of data can be transformed using hashing techniques into a digital hash or fixed-length fingerprint. A digital hash cannot be reversed by a criminal to reveal the original input. A different hash is produced whenever there is any change in the input. Password protection is achieved in this way. In order to protect passwords and ensure that users are using the proper ones, systems must store passwords in a secure manner.

3.0.4 Preventing Criminals



For an attacker to decode a hash, they must guess the password. When it comes to password guessing, brute-force and dictionary assaults are the most widely employed techniques.

In a dictionary attack, a file comprising widely used words, phrases, and passwords is employed. The file contains the hashes that are computed. A dictionary attack compares the hashes of the password with the document copies. If the hash contents match, the attacker will be aware of a potential set of strong passwords.

Up to a predetermined length, every character combination is tried in a brute-force attack. Even though a brute-force method uses a lot of computer power, it will eventually find the password. A password needs to be adequate in length so that it would take a while to crack.

In-Text Question (ITQ)

What are some examples of technopreneurship initiatives in **cybersecurity**?

In-Text Answer (ITA)

Examples of technopreneurship initiatives in **cybersecurity** include startups developing **AI**-powered threat detection systems, companies offering secure authentication solutions based on biometrics or blockchain technology, and ventures focusing on secure software development practices such as DevSecOps. These initiatives aim to address specific **cybersecurity** challenges and provide innovative solutions to protect organisations from cyber threats. There are no simple solutions that address the moral dilemmas that **cybersecurity** professionals encounter. Depending on the circumstances and the ethical viewpoint, the response to the question of what is right or wrong can vary.

3.1 A Digital Signature: What Is It?

Stamped seals and handwritten signatures attest to the authorship of a document's contents. The usefulness of handwritten signatures can be matched by digital signatures.



In a number of countries, e-signatures are legally equivalent to handwritten signatures.

Digital documents that are not safeguarded are incredibly simple for anyone to alter. If a document is edited after the user signs it, it can be detected by a digital signature. An electronic signature provides a mathematical technique that guarantees accuracy and quality in the digital document, software, or transmission.

In a number of countries, e-signatures are legally equivalent to handwritten signatures. Contracts, discussions, and any other document requiring a handwritten signature are binding when signed electronically. The history of the electronic document is tracked by an audit trail for legal and regulatory defence needs.

Authenticity, integrity, and non-repudiation are established with the use of a digital signature. As seen in the picture, some characteristics of digital signatures allow for data integrity and entity authentication.

3.1.1 Procedures for Digital Signature Creation

Digital signatures use asymmetric cryptography as their foundation. A common key solution generates a pair of keys: a secret key and an open key. It has a mathematical connection to the credentials. [Go to this URL](#) for more information about digital signatures.

3.1.2 Data Accuracy

Data can be efficiently stored, retrieved, and analysed with the help of databases. Professionals in **cybersecurity** must safeguard the expanding quantity of databases as data collecting grows and becomes more sensitive. Consider a database as a system of electronic files. The management of the company, database designers, and developers are accountable for maintaining data integrity.

3.2 The Idea of the Five Nines

Enterprises seeking to optimise system and data availability may need to take drastic steps to reduce or completely eradicate data loss. Reducing mission-critical process downtime is the aim. If workers are unable to do their



assigned tasks, the company may experience a loss of income.

Organisations use uptime as a percentage to gauge availability. We start this chapter by going over the idea of five nines. The highest availability standards are required in many businesses, where downtime could actually mean the difference between life and death.

In order to guarantee the availability of computers or network systems, redundancy adds additional components and backup. Redundant components can include software as well as equipment, such as desktops, laptops, apps, and information systems, as well as storage devices, server infrastructure, and intermediary devices. By defining protocols that they adhere to following an event, organisations can better prepare themselves to handle an incident.

3.2.1 What Signifies the Five Nines?

- A five-nine rating indicates that services and systems are operational 99.999% of the time. Additionally, it indicates that the annual downtime, both scheduled and unexpected, is less than 5.26 minutes. The figure's chart compares the downtime for different availability percentages.
- A system or component with high availability is one that is up and running all the time for a predetermined amount of time.

To assist in guaranteeing high availability:

- Get rid of isolated sites of failure
- Build with dependability in mind
- See failures early on

Maintaining high availability at the five-nine standard can be expensive and resource-intensive. The purchase of extra gear, including servers and components, is the cause of the cost increases. The complexity of configuration rises when an organisation adds more components. Regrettably, the risk



variables rise with configuration complexity. The chance of a component failing increases with the number of moving parts.

3.2.2 Conditions That Call for Five Nines

- In the finance sector, high availability is essential to continued trading
- compliance and client trust
- In order for healthcare facilities to provide patients with round-the-clock care, high availability is required.
- Organizations that offer security and services to a town, state, or country are included in the public safety industry
- Product distribution to customers and effective supply chains are critical to the retail industry
- Disruption can be disastrous, particularly during periods of high demand, like holidays
- The public expects the news media to report events as they happen. These days, the news cycle runs continuously, nonstop
- A successful denial-of-service attack substantially disrupts operations
- A substantial amount of sensitive data is lost by an organisation
- An essential application stops working
- An administrator or root user is compromised.
- The identification of an illicit file server shares or cross-site script
- Public relations are impacted when an organisation's website is vandalised.
- A powerful storm, like a hurricane or tornado; a disastrous incident, like a building explosion, terrorist attack, or fire; an extended utility or service provider outage; water damage from flooding or sprinkler failure
- An organisation can realise a threat's financial impact by classifying each threat's impact level. To view an example of each threat category

**In-Text Question (ITQ)**

Explain the concept of social engineering in the context of penetration testing.

In-Text Answer (ITA)

Social engineering involves manipulating individuals to divulge confidential information or perform actions that may compromise security. In penetration testing, social engineering techniques are often used to gather information, gain access to restricted areas, or trick employees into providing sensitive information such as passwords.

3.3 Creating a High Availability System

Three main ideas are combined in high availability to accomplish the objective of continuous access to data and services:

1. Single points of failure are eliminated or reduced
2. Sturdiness of the System
3. Tolerance for Faults

Understanding how to manage one instance of failure is essential. Information technology Infrastructure, centralised intermediary devices, and even highly skilled computer professionals can all be a single point of failure. The important point is that losing a person, system, or process can have a seriously disruptive effect on the system as a whole. It is essential to have parts, facilities, and processes whose services prevent an instant disaster. Redundancy can be achieved, for example, by high availability clusters. These clusters are made up of several computers with identical network setups and access to the same shared storage. Every server processes a service concurrently. When seen externally, the server group appears to be a single device. The remaining servers in the cluster keep processing the same service as the failed device in the event of a server failure.

The ability of a system to continue to process data and remain functioning in the face of disruptions or attacks is



known as its resilience. This usually calls for redundant systems—that is, systems with enough processing and power to take over in the event of a system failure without causing a service interruption. More than just hardening equipment, system resilience necessitates the availability of data and services even in the face of assault.

- A system with fault tolerance can function even in the event that one or more components malfunction. Fault tolerance is exemplified by data mirroring. In the event that a "fault" arises that disrupts a device, like a disk controller, the mirrored system gives the user access to the required data without appearing to disturb service.
- Examination of Risks
- The process of examining the risks that both naturally occurring and man-made events pose to an organisation's assets is known as risk analysis.
- To help decide which assets need to be protected, a user does an asset identification.

Three objectives define a risk analysis:

- Determine the worth of the assets
- Identify threats and vulnerabilities
- Weigh an expense associated with the countermeasure with the danger it poses.

3.3.1 Measure Chances And Effect Of The Risks

A team evaluates each threat to an asset and plots it in the table. The team ranks the results and uses the results as a guide. They may decide to take action only on threats that fall within the red zone.

3.3.2 Risk Reduction

Reduction of Mitigation is the process of lessening the extent of the loss or the probability that it will occur. Numerous technical restrictions, such as firewalls, file permissions, and authentication systems, reduce risk. Professionals in organisation and security need to be aware that risk



reduction can affect an organisation in both positive and negative ways. Effective Hazard mitigation achieves a state of equilibrium between the advantages of risk reduction and the drawbacks of controls and countermeasures.

There are four typical strategies to lower risk:

- Assign this danger to an outside group
- Accept the risk and occasionally re-evaluate
- Implement controls to reduce the risk
- Completely alter the method to avoid the risk

Taking on the risk that demands the development of backup plans is a short-term tactic. Every day, individuals and businesses must embrace risk by gradually creating software and regularly releasing updates and patches to fix bugs and misconfigurations, and modern approaches lower risk.

Purchasing insurance, maintenance contracts, or outsourcing services are a few instances of risk transfer. It can be a wise move to hire experts to handle crucial duties in order to lower risk and produce better outcomes with shorter investment periods. Two or more strategies can be included in an effective risk reduction plan.

3.3.3 N+1 Redundancy

In the case of a component failure, system availability is guaranteed via N+1 redundancy. There must be a minimum of one backup component (+1) for components (N). In the case of a flat tyre, an automobile, for instance, has four tyres (N) and an additional tyre in the trunk (+1).

N+1 redundancy in a data centre relates to the device's functionality architecture to survive the loss of a component. The data centre's numerous constituents, such as servers, power supplies, switches, and routers, are collectively referred to by the letter N. The extra part or system that is ready to go when needed is represented by the +1.

A data centre's power generator, which kicks in when the primary power supply fails, is an example of N+1 redundancy.



An N+1 system is not completely redundant, even though it has redundant hardware.

- Systems for Detecting Intrusions
- The traffic on a network is passively observed by intrusion detection systems or IDSs.

When an IDS is passive, it keeps track of and reports on traffic. It doesn't do anything at all. This is how working in a promiscuous mode is defined.

The benefit of using a copy of the traffic is that the packet flow of the forwarded traffic is not adversely impacted by the intrusion detection system. Using a duplicate of the traffic has the drawback which malicious single-packet assaults cannot be prevented by the IDS from reaching the target before the attack is countered. Using a gadget that can quickly identify and thwart an assault is a better option. This is done via an intrusion prevention system (IPS).

3.3.4 Testing for Penetration

Using a variety of malevolent tactics, penetration testing, often known as pen testing, is a technique for identifying system vulnerabilities. **Vulnerability** testing and pen testing are two different things. Testing for vulnerabilities merely points out possible issues. Pen testing entails a **cybersecurity** expert breaking into a website, network, or server with authorisation from the enterprise in an attempt to access resources using credentials, usernames, or other common information. The primary distinction between **cybersecurity** experts and cybercriminals is that the latter are authorised by the organisation to carry out the examinations.

One of the main goals of pen testing for a business is to identify and address vulnerabilities before cybercriminals do. Another name for penetration testing is hacking with ethics. Packet analysers.

Net traffic is intercepted and logged by packet analysers, sometimes known as packet sniffers. Every packet is captured by the packet analyser, which also examines and evaluates



its contents. Network traffic on wired and wireless networks can be captured by a sniffer.

Packet analysers carry out the subsequent tasks:

- Analysing network issues;
- Spotting efforts at network intrusion

Traffic recording;

- System isolation for vulnerable systems;
- Network misuse detection

In-Text Question (ITQ)

Question for the lab exam: How do you set up a Kali Linux cybersecurity lab for ethical hacking as a technopreneurship response to threats?

In-Text Answer (ITA)

- **Install Virtualization Software:** You'll need virtualisation software such as VMware Workstation, Virtual Box, or VMware Fusion (for **Mac**) to create virtual **machines** (VMs). Choose one and install it on your host system.
- **Download Kali Linux ISO:** Visit the official Kali Linux website and download the ISO file for the version you prefer. You can choose between Kali Linux (64-bit) or Kali Linux (32-bit), depending on your system architecture.
- **Create a Virtual Machine:** Open your virtualisation software and create a new virtual machine. During the creation process, specify the Kali Linux ISO file you downloaded as the installation media.
- **Configure Virtual Machine Settings:** Adjust the virtual machine settings as needed. This includes specifying the amount of RAM, CPU cores, and hard disk space allocated to the VM.
- **Install Kali Linux:** Start the virtual machine and follow the on-screen instructions to install Kali Linux. You can choose either a graphical or text-based installation, depending on your preference.



- **Update and Upgrade Kali Linux:** After installation, open a terminal window in Kali Linux and run the following commands to update the package repository and upgrade installed packages: **sudo apt update** and **sudo apt upgrade**
- **Install Penetration Testing Tools:** Kali Linux comes pre-installed with a wide range of penetration testing tools. You can install additional tools using the apt package manager. For example, to install the Metasploit Framework, you can use the following command: **sudo apt install Metasploit-framework**
- **Set Up Networking:** Configure networking for your virtual lab. You can choose from different networking options, such as NAT, Bridged, or Host-only networking, depending on your requirements.
- **Create Target Machines:** Set up additional virtual machines to act as target systems for your penetration testing exercises. You can install vulnerable operating systems such as Windows XP, Windows 7, or older versions of Linux distributions.
- **Practice Penetration Testing:** Once your lab is set up, you can start practising penetration testing techniques using the tools available in Kali Linux. There are numerous online resources, tutorials, and labs available to guide you through various penetration testing scenarios.
- **Document Your Findings:** As you practice, make sure to document your findings, including vulnerabilities discovered, exploitation techniques used, and remediation recommendations. This documentation will help you reinforce your learning and track your progress.
- **Stay Updated: Cybersecurity** is a constantly evolving field, so it's essential to stay updated with the latest tools, techniques, and vulnerabilities. Follow relevant



blogs, forums, and news sources to stay informed about the latest developments in penetration testing and **cybersecurity**.

3.3.5 Tools for Security

When it comes to the greatest security technologies, numerous factors will rely on the context, circumstances, and individual preferences. An expert in **cybersecurity** needs to know where to go for reliable information.

Kali software is used by IT specialists to assess the security of their networks. On a Linux platform, Kali Linux integrates more than 300 security audits and penetration testing tools.

- Situational Awareness of Networks
- An organisation must be able to keep an eye on networks, assess the information that comes in, and spot hostile activities. To access a variety of CERT-developed traffic analysis tools, see this link.
- Specifying the Functions of Experts in **Cybersecurity**
- Professionals in **cybersecurity** have a defined job according to ISO standards.

Information security experts, security architects, site/physical security managers, facilities contacts, HR representatives for HR-related issues like training and disciplinary action, systems and network managers, and other IT specialists

Breakdown of the several kinds of information security positions:

- Definers include consultants who perform risk assessments, build technological architectures and products, and give rules, guidelines, and standards. They also include senior-level employees of organisations that have a broad but limited depth of expertise.
- The true techies who design and implement security solutions are builders; they monitor and manage security equipment, carry out security monitoring duties and enhance procedures.



3.3.6 Tools for Job Searches

Information technology jobs are advertised on a multitude of websites and mobile applications. Every website caters to a distinct set of job seekers and offers various resources to help them find their dream job. Many websites function as job site marketplaces, which are essentially recruitment search engines that advertise openings from numerous corporate sites for employment and provide them in one location.

- **True.com**

Claimed to be the most popular employment site globally, Indeed.com welcomes more than 180 million unique visits from more than 50 nations each month. Indeed, it is a genuinely global job platform. Indeed, provides the best opportunity for job searchers and assists businesses of all sizes in hiring the most talented individuals.

- **CareerBuilder.com**

CareerBuilder provides services to numerous sizable, esteemed firms. Consequently, candidates with greater education and qualifications tend to be drawn to this site. Companies using CareerBuilder often receive a higher volume of applications from people with advanced degrees, industry certifications, and advanced credentials.

3.4 Summary

Overall, technopreneurship in **cybersecurity** is essential for addressing the ever-evolving cyber threats faced by individuals, businesses, and governments. By combining technological innovation with entrepreneurial vision, technopreneurs play a critical role in safeguarding digital assets and promoting a secure cyberspace.

3.5 Conclusion

Technology serves as a cornerstone for technopreneurship development in social media marketing and **cybersecurity**, empowering entrepreneurs to create value, mitigate risks, and capitalise on emerging trends. By leveraging technology effectively, technopreneurs can drive growth, foster innovation, and build resilient businesses in an ever-evolving



digital.

The convergence of technology, entrepreneurship, and innovation in social media marketing and **cybersecurity** represents a paradigm shift in how businesses operate and defend themselves in the digital age. By harnessing the power of technology and adopting a proactive entrepreneurial mindset, technopreneurs are not only reshaping industries but also shaping the future of commerce and security in an increasingly interconnected world landscape.

3.6 Self-Assessment Exercise

1. What is the primary purpose of a code of ethics for **cybersecurity** experts?
2. What are some key principles typically included in a code of ethics for **cybersecurity** experts?
3. What role does accountability play in **cybersecurity** ethics?

3.7 Tutor-Marked Assignment

Submission of final report:

- Teams should submit a final written report that documents their proposal in **detail**.
- The exercise's success and actions should be documented and submitted to Dropbox to ensure you have developed sufficient capacity to effectively impart your knowledge.



11.0 SELF-ASSESSMENT ANSWERS

MODULE ONE

11.1.1 MODULE 1: SELF-ASSESSMENT ANSWERS (UNIT 1)

1. Challenges include rapidly evolving technology landscapes requiring continuous learning and adaptation, **cybersecurity** threats, funding constraints for technology development, regulatory compliance issues, and the need to navigate complex intellectual property rights.
2. Governments and institutions can support technopreneurship by providing funding and grants for technology research and development, fostering collaborative ecosystems between academia, industry, and startups, offering tax incentives for innovation-driven businesses, and implementing supportive policies for intellectual property protection and entrepreneurship education.

MODULE TWO

11.2.1 MODULE 2: SELF-ASSESSMENT ANSWERS (UNIT 2)

1. Data analytics plays a crucial role in social media marketing for technopreneurs by providing insights into the effectiveness of their marketing campaigns, audience demographics and preferences, engagement metrics, and **ROI**. Technopreneurs can use data analytics to track key performance indicators, identify trends, and make informed decisions to optimise their social media marketing strategies for better results.
2. Technopreneurs can leverage social media to drive sales and conversions by implementing strategies such as showcasing product features and benefits, offering exclusive promotions or discounts to followers, providing seamless purchasing options through integrated e-commerce platforms, utilising social proof through customer testimonials and reviews, and retargeting interested users with personalised offers based on their



interactions with the brand on social media.

MODULE THREE

11.3.1 MODULE 3: SELF-ASSESSMENT ANSWERS (UNIT 3)

1. Companies can utilise social media platforms to promote their products or services, engage with customers, and increase brand visibility through targeted advertising campaigns and content marketing strategies.
2. Professionals can leverage social media platforms like LinkedIn to connect with industry peers, participate in relevant groups or communities, and stay updated on industry trends and news.

MODULE FOUR

11.4.1 MODULE 4: SELF-ASSESSMENT ANSWERS (UNIT 4)

1. Social media marketing can benefit your business by increasing brand awareness, allowing targeted advertising, improving customer engagement, offering cost-effective marketing options, and providing valuable customer insights.
2. Drawbacks include the time-consuming nature of managing accounts and creating content, the potential amplification of negative feedback, dependency on platform algorithms, measurement challenges in determining **ROI**, and the risk of oversaturation in a competitive landscape.

MODULE SIX

11.5.1 MODULE 6: SELF-ASSESSMENT ANSWERS (UNIT 6)

1. **Encryption:** Use encryption to protect sensitive data both in transit and at rest. This ensures that even if the data is intercepted, it cannot be read without the appropriate decryption keys.
2. **Strong Passwords:** Enforce the use of strong, complex



passwords or passphrases for accessing systems and accounts. Encourage regular password changes and discourage the reuse of passwords across multiple accounts.

3. **Multi-Factor Authentication (MFA):** Implement MFA wherever possible. This adds an extra layer of security by requiring users to provide multiple forms of identification to access a system or application.
4. **Regular Updates and Patch Management:** Keep all systems, software, and devices up to date with the latest security patches and updates. Vulnerabilities in software can often be exploited by attackers, so staying current is crucial.
5. **Firewalls and Intrusion Detection Systems (IDS):** Use firewalls to monitor and control incoming and outgoing network traffic. IDS can help detect and respond to suspicious activity on your network.
6. **Access Control:** Limit access to sensitive information to only those who need it to perform their job duties. Use role-based access control (RBAC) to ensure that users only have access to the resources necessary for their roles.
7. **Data Backups:** Regularly backup important data and ensure that backups are stored securely off-site. This helps mitigate the impact of data loss due to cyberattacks or other incidents.
8. **Employee Training and Awareness:** Educate employees about **cybersecurity** best practices and the potential risks of cyber threats such as phishing and social engineering attacks. Encourage a culture of security awareness within the organisation.
9. **Incident Response Plan:** Develop and regularly test an incident response plan to ensure a swift and effective response to security incidents. This plan should outline procedures for detecting, responding to, and recovering from cyberattacks.



10. **Regular Security Audits and Assessments:** Conduct regular security audits and assessments to identify and address any vulnerabilities or weaknesses in your systems and processes.

By implementing these measures and staying vigilant, organisations can significantly reduce the risk of data breaches and keep their information safe in the face of evolving **cybersecurity** threats.

MODULE SEVEN

11.6.1 MODULE 7: SELF-ASSESSMENT ANSWERS (UNIT 7)

1. **Encryption:** Use encryption to protect sensitive data both in transit and at rest. This ensures that even if the data is intercepted, it cannot be read without the appropriate decryption keys.
2. **Strong Passwords:** Enforce the use of strong, complex passwords or passphrases for accessing systems and accounts. Encourage regular password changes and discourage the reuse of passwords across multiple accounts.
3. **Multi-Factor Authentication (MFA):** Implement MFA wherever possible. This adds an extra layer of security by requiring users to provide multiple forms of identification to access a system or application.
4. **Regular Updates and Patch Management:** Keep all systems, software, and devices up to date with the latest security patches and updates. Vulnerabilities in software can often be exploited by attackers, so staying current is crucial.
5. **Firewalls and Intrusion Detection Systems (IDS):** Use firewalls to monitor and control incoming and outgoing network traffic. IDS can help detect and respond to suspicious activity on your network.
6. **Access Control:** Limit access to sensitive information to only those who need it to perform their job duties. Use role-



based access control (RBAC) to ensure that users only have access to the resources necessary for their roles.

7. **Data Backups:** Regularly backup important data and ensure that backups are stored securely off-site. This helps mitigate the impact of data loss due to cyberattacks or other incidents.
8. **Employee Training and Awareness:** Educate employees about **cybersecurity** best practices and the potential risks of cyber threats such as phishing and social engineering attacks. Encourage a culture of security awareness within the organisation.
9. **Incident Response Plan:** Develop and regularly test an incident response plan to ensure a swift and effective response to security incidents. This plan should outline procedures for detecting, responding to, and recovering from cyberattacks.
10. **Regular Security Audits and Assessments:** Conduct regular security audits and assessments to identify and address any vulnerabilities or weaknesses in your systems and processes.

By implementing these measures and staying vigilant, organisations can significantly reduce the risk of data breaches and keep their information safe in the face of evolving **cybersecurity** threats.

MODULE 8

11.7.1 MODULE 8: SELF-ASSESSMENT ANSWERS (UNIT 8)

ANSWER 1

The primary purpose of a code of ethics for **cybersecurity** experts is to provide guidelines and principles that govern their behaviour and decision-making processes in the field. It helps ensure that **cybersecurity** professionals act responsibly, ethically, and with integrity to protect the confidentiality, integrity, and availability of data and systems.

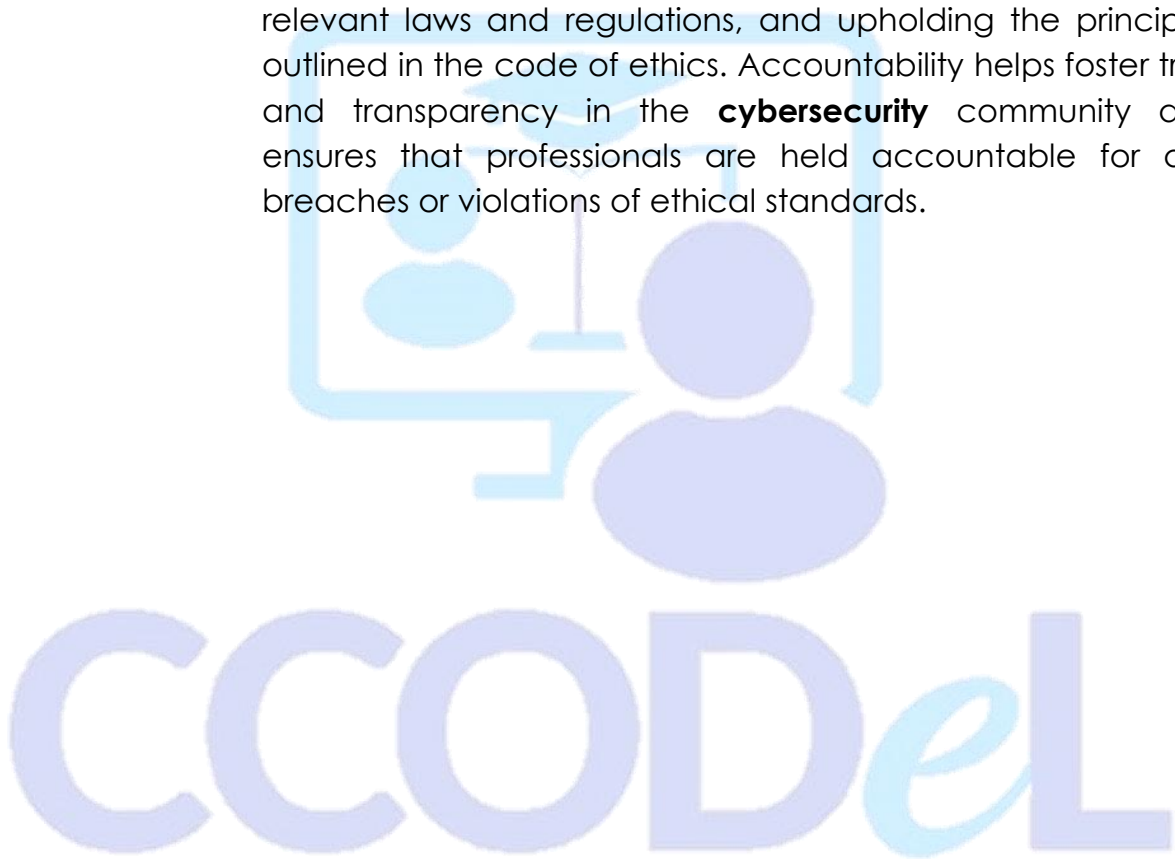
ANSWER 2



Some key principles commonly included in a code of ethics for **cybersecurity** experts include confidentiality, integrity, availability, accountability, professionalism, competence, and respect for privacy and individual rights.

ANSWER 3

Accountability is essential in **cybersecurity** ethics as it involves taking responsibility for one's actions and decisions. **Cybersecurity** professionals are accountable for ensuring the security of systems and data under their care, adhering to relevant laws and regulations, and upholding the principles outlined in the code of ethics. Accountability helps foster trust and transparency in the **cybersecurity** community and ensures that professionals are held accountable for any breaches or violations of ethical standards.





12.0 GLOSSARY

VULNERABILITY - A vulnerability is a weakness that hackers could use to enter a computer system without authorisation.

CYBERSECURITY - Cybersecurity seeks to defend against computer viruses, complex and expensive ransomware attacks, and other threats to people's and organisations' systems, apps, computing devices, sensitive data, and financial assets.

BUSINESS-SAVVY - A business issue (risks and possibilities) should be understood and handled quickly and keenly in a way that will most likely result in a positive conclusion. This is known as business expertise and is sometimes referred to as business savviness, business sense, and business knowledge.

IoT - The Internet of Things, or IoT, is the collective term for the network of interconnected gadgets as well as the technology that enables communication between devices and the cloud.

SWOT - SWOT Analysis: What Is It? A technique for determining and evaluating internal strengths and weaknesses as well as external opportunities and threats that influence present and future operations and aid in the development of strategic goals is called SWOT analysis (strengths, weaknesses, opportunities, and threats).

KPIs - Key performance indicators, or KPIs for short, are quantitative measurements of performance over time toward a certain goal. KPIs offer goals for teams to strive towards, benchmarks to assess advancement, and perspectives to assist everyone throughout the company in making more informed choices.

CEO - Key Takeaways. The chief executive officer (**CEO**) is **the highest-ranking person in a company**. While every company differs, **CEOs** are often responsible for expanding the company, driving profitability, and, in the case of public companies, improving share prices. **CEOs** manage the overall operations of a company.

Gen Z - According to the Merriam-Webster Online Dictionary, Generation Z is "the generation of people born in the late 1990s and early 2000s." Generation Z is defined by the Collins Dictionary as "those who were born between the middle of the 1990s and the middle of the 2010s."



X - The Social Media Platform Previously Known as Twitter's New Name and Logo

ROI - An investor can use the mathematical formula known as return on investment, or **ROI**, to assess their investments and determine how well a specific investment has done in comparison to others.

SMART - Setting goals is **aided** by the usage of SMART goals. The words "specific, measurable, achievable, realistic, and timely" are abbreviated as "SMART."

AI - Artificial intelligence (**AI**) is the capacity of a computer or robot under computer control to carry out operations typically performed by intelligent entities.

NGOs - Introduced for the first time in 1945, non-governmental organisations, or NGOs, are non-profit organisations free from the influence of the government, but they are sometimes funded by it.

BOTNET - A **botnet** is a group of infected computers controlled by a malicious actor, with each infected device called a bot.

MAC - A **MAC** address is a unique 12-character alphanumeric attribute used to identify individual electronic devices on a network, such as 00-B0-D0-63-C2-26.

ARP - Address Resolution Protocol (**ARP**) is a protocol that connects an evolving IP address to a fixed **MAC** address in a local-area network (LAN).

IP - An IP address is a numerical label assigned to a computer network device, serving two primary functions: network interface identification and location addressing.

DevSecOps - DevSecOps is an IT approach that emphasises development, security, and operations, integrating security as a shared responsibility throughout the entire IT lifecycle.

ATM - An ATM is an automated teller **machine** that allows you to withdraw cash from your bank account without physically visiting a teller.

HASHING - Hashing is the conversion of a key or string into a new value, commonly used in hash tables to simplify the retrieval of the original string.